

STRUCTURAL ANALYSIS AND DESIGN REPORT

Pelican House, Exuma, Bahamas

Project: A-1599

May 19th, 2026

Prepared By

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1. Structural Framing Analysis and Design

The structural calculations for Pelican House, Exuma, Bahamas were performed using FORTEWEB and Tekla Tedds. Structural Analysis and desing for the Swimming Pool was done using Tekla Structural Designer. The analysis was conducted in accordance with applicable provisions of ASCE 7, ACI 318-19, TMS 402/602, IRC 2021, NDS and the governing Bahamas Building Code requirements.

Design loads were considered as follows:

1. Dead Load — Applied based on member self-weight, CMU wall weight, roof dead load, and other permanent loads.
2. Roof Live Load — Applied as required for residential roof framing.
3. Snow Load — Taken as 0 psf due to the project location.
4. Wind Load — Based on $V=180$ mph (as per Dlubal Geozone Tool).
5. Seismic Load — Considered non-governing for the local masonry wall and footing checks compared with wind loading.

Calculations for the analyzed framing members, masonry walls, columns, footings, and supporting elements are attached as Annexure A. Design summary and calculations for the Swimming Pool are attached as Annexure B.

2. Lateral Analysis

A simplified lateral analysis was performed for the proposed Pelican House, Exuma, Bahamas structure in accordance with applicable provisions of ASCE 7, TMS 402/602, and the governing Bahamas Building Code requirements. The structure consists primarily of 8-inch CMU load-bearing exterior walls, with openings provided in selected wall elevations.

Wind loading was evaluated using a basic wind speed of 180 mph, Risk Category II, Exposure C, enclosed building classification, and a calculated velocity pressure of $q_h=60.67$ psf, based on the project wind analysis.

The critical masonry wall panel was identified from the exterior wall elevations based on the wall opening layout and available solid wall pier lengths. The selected critical CMU wall panel was checked for both in-plane shear demand and out-of-plane wind pressure. Out-of-plane wall loading was evaluated using the controlling wall pressure from the wind analysis, while in-plane shear resistance was reviewed based on the available masonry shear wall panel geometry and load path assumptions.

The masonry wall panels, local columns, and supporting footing elements were analyzed using Tekla Tedds and supporting hand calculations. Based on the completed checks, the critical masonry wall panel and associated support elements were found adequate for the applied wind and gravity loading conditions.

3. Footing Analysis

Foundation checks were performed for the proposed Pelican House, Exuma, Bahamas structure in accordance with applicable provisions of ACI 318-19, TMS 402/602, ASCE 7, and the governing Bahamas Building Code requirements.

The foundation system was reviewed for the applied gravity and lateral load effects from the CMU walls, roof framing, local columns, and deck post supports. The following footing elements were analyzed:

- Strip footing supporting the 8-inch CMU load-bearing masonry walls.
- Isolated pad footing supporting the local column carrying roof beam reactions.
- Deck post footings supporting the exterior deck framing/post loads.

The strip footing was checked for wall axial load, footing self-weight, soil bearing, and wind-related lateral effects from the masonry wall system. The column footing was checked for the concentrated axial load transferred from the supported roof beams and column self-weight. The deck post footings were checked for the applied post reactions from the exterior deck framing.

Based on the completed footing checks, the strip footing, column footing, and deck post footings were found adequate for the applied loading conditions. Detailed footing calculations are included in Annexure A.

4. Conclusion

The proposed Pelican House, Exuma, Bahamas structure was analyzed in accordance with applicable provisions of ASCE 7, ACI 318-19, TMS 402/602, IRC 2021, NDS and the governing Bahamas Building Code requirements. Structural framing members, CMU wall panels, local columns, strip footings, column footings, and deck post footings were reviewed for the applicable gravity, wind, and support reactions.

Based on the completed structural analysis, the critical masonry wall panels and foundation elements were found adequate for the applied loading conditions. Therefore, the proposed structure is considered structurally adequate and acceptable from a structural engineering standpoint.

Level			
Member Name	Results (Max UTIL %)	Current Solution	Comments
Roof Rafters	Passed (62% M)	1 piece(s) 2 x 8 DF No.1 @ 16" OC	
Roof: Hip/Valley Beam	Passed (94% M)	3 piece(s) 2 x 12 DF No.1	
Deck: Floor Joists	Passed (78% M)	1 piece(s) 2 x 8 DF No.1 @ 16" OC	
Deck Beams	Passed (93% M)	1 piece(s) 2 x 8 DF No.1	
Deck Posts	Passed (8% f _c)	1 piece(s) 4 x 4 DF No.1	

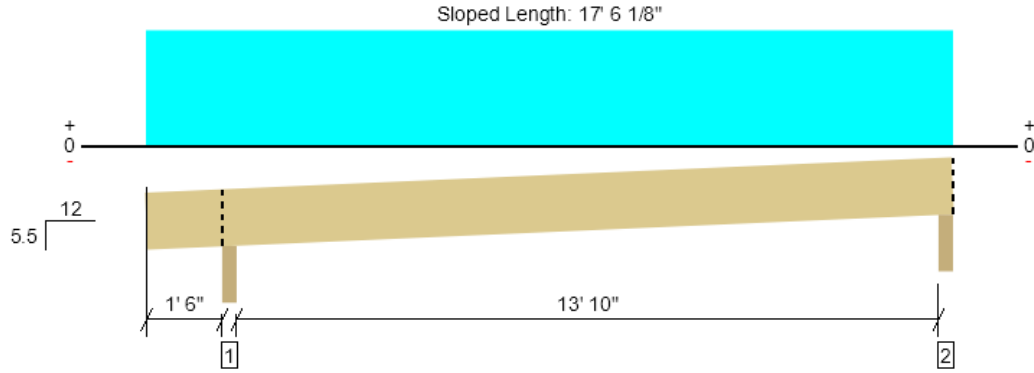
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Level, Roof Rafters

1 piece(s) 2 x 8 DF No.1 @ 16" OC



Drawing is Conceptual. All locations are measured from the outside face of left support (or left cantilever end). All dimensions are horizontal (typ.).

Design Results	Actual @ Location	Allowed	Result	LDF	Load: Combination (Pattern)
Member Reaction (lbs)	427 @ 1' 7 3/4"	2454 (3.50")	Passed (17%)	--	1.0 D + 1.0 Lr (All Spans)
Shear (lbs)	313 @ 2' 4 1/16"	1631	Passed (19%)	1.25	1.0 D + 1.0 Lr (All Spans)
Moment (Ft-lbs)	1179 @ 8' 8 15/16"	1889	Passed (62%)	1.25	1.0 D + 1.0 Lr (Alt Spans)
Live Load Defl. (in)	0.345 @ 8' 8 5/16"	0.773	Passed (L/538)	--	1.0 D + 1.0 Lr (Alt Spans)
Total Load Defl. (in)	0.625 @ 8' 8 3/8"	1.031	Passed (L/297)	--	1.0 D + 1.0 Lr (Alt Spans)

- Deflection criteria: LL (L/240) and TL (L/180).
- Overhang deflection criteria: LL (2L/240) and TL (2L/180).
- Allowed moment does not reflect the adjustment for the beam stability factor.
- A 15% increase in the moment capacity has been added to account for repetitive member usage.
- Applicable calculations are based on NDS.

Member Length : 17' 9 7/16"
System : Roof
Member Type : Joist
Building Use : Residential
Building Code : IBC 2024
Design Methodology : ASD
Member Pitch : 5.5/12

Supports	Bearing Length			Loads to Supports (lbs)			
	Total	Available	Required	Dead	Roof Live	Factored	Accessories
1 - Beveled Plate - SPF	3.50"	3.50"	1.50"	193	234	427	Blocking
2 - Beveled Plate - SPF	3.50"	3.50"	1.50"	157	192	349	Blocking

• Blocking Panels are assumed to carry no loads applied directly above them and the full load is applied to the member being designed.

Lateral Bracing	Bracing Intervals	Comments
Top Edge (Lu)	9' o/c	
Bottom Edge (Lu)	17' 6" o/c	

- Maximum allowable bracing intervals based on applied load.
- Dimensions for lateral bracing intervals are measured along the length of the member for sloped conditions.

Vertical Load	Location (Side)	Spacing	Dead (0.90)	Roof Live (1.25)	Comments
1 - Uniform (PSF)	0 to 15' 11"	16"	15.0	20.0	Default Load

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The product application, input design loads, dimensions and support information have been provided by ForteWEB Software Operator

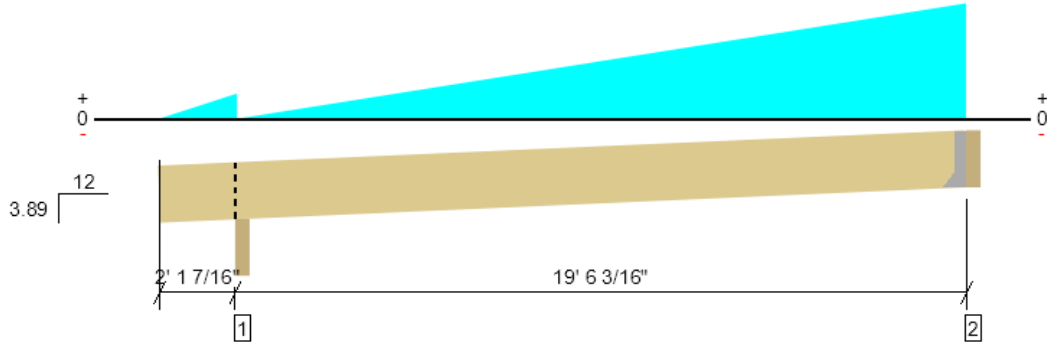
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Level, Roof: Hip/Valley Beam
3 piece(s) 2 x 12 DF No.1

Sloped Length: 23' 5/8"



Drawing is Conceptual. All locations are measured from the outside face of left support (or left cantilever end). All dimensions are horizontal (typ.).

Design Results	Actual @ Location	Allowed	Result	LDF	Load: Combination (Pattern)
Member Reaction (lbs)	2264 @ 21' 7 5/8"	4219 (1.50")	Passed (54%)	--	1.0 D + 0.7 S (Alt Spans)
Shear (lbs)	1965 @ 20' 8 15/16"	6986	Passed (28%)	1.15	1.0 D + 0.7 S (Alt Spans)
Moment (Ft-lbs)	8577 @ 13' 4 1/4"	9097	Passed (94%)	1.15	1.0 D + 0.7 S (Alt Spans)
Live Load Defl. (in)	0.332 @ 12' 3 13/16"	1.018	Passed (L/737)	--	1.0 D + 0.7 S (Alt Spans)
Total Load Defl. (in)	0.690 @ 12' 3 1/2"	1.358	Passed (L/354)	--	1.0 D + 0.7 S (Alt Spans)

- Deflection criteria: LL (L/240) and TL (L/180).
- Overhang deflection criteria: LL (2L/240) and TL (2L/180).
- Allowed moment does not reflect the adjustment for the beam stability factor.
- Applicable calculations are based on NDS.

Member Length : 23' 5/8"
System : Roof
Member Type : Flush Beam
Building Use : Residential
Building Code : IBC 2024
Design Methodology : ASD
Member Pitch : 3.89/12

Supports	Bearing Length			Loads to Supports (lbs)			Accessories
	Total	Available	Required	Dead	Snow	Factored	
1 - Beveled Plate - SPF	3.50"	3.50"	1.50"	723	858	1324	Blocking
2 - Hanger on 11 1/4" SPF beam	3.50"	Hanger ¹	1.50"	1158	1580	2264	See note ¹

- Blocking Panels are assumed to carry no loads applied directly above them and the full load is applied to the member being designed.
- At hanger supports, the Total Bearing dimension is equal to the width of the material that is supporting the hanger
- ¹ See Connector grid below for additional information and/or requirements.

Lateral Bracing	Bracing Intervals	Comments
Top Edge (Lu)	6' 4" o/c	
Bottom Edge (Lu)	22' 9" o/c	

- Maximum allowable bracing intervals based on applied load.
- Dimensions for lateral bracing intervals are measured along the length of the member for sloped conditions.

Connector: Simpson Strong-Tie

Support	Model	Seat Length	Top Fasteners	Face Fasteners	Member Fasteners	Accessories
2 - Face Mount Hanger	HU210-3X SLD17	2.50"	N/A	18-16d	10-10d	

- Refer to manufacturer notes and instructions for proper installation and use of all connectors.

Vertical Loads	Location (Side)	Tributary Width	Dead (0.90)	Snow (1.15)	Comments
0 - Self Weight (PLF)	0 to 21' 7 5/8"	N/A	12.8	--	
1 - Tapered (PLF)	0 to 2' 1 7/16"	N/A	0.0 to 33.3	0.0 to 53.0	Generated from Roof Geometry
2 - Tapered (PLF)	2' 1 7/16" to 21' 7 5/8"	N/A	0.0 to 151.4	0.0 to 244.0	Generated from Roof Geometry

- Side loads are assumed to not induce cross-grain tension.

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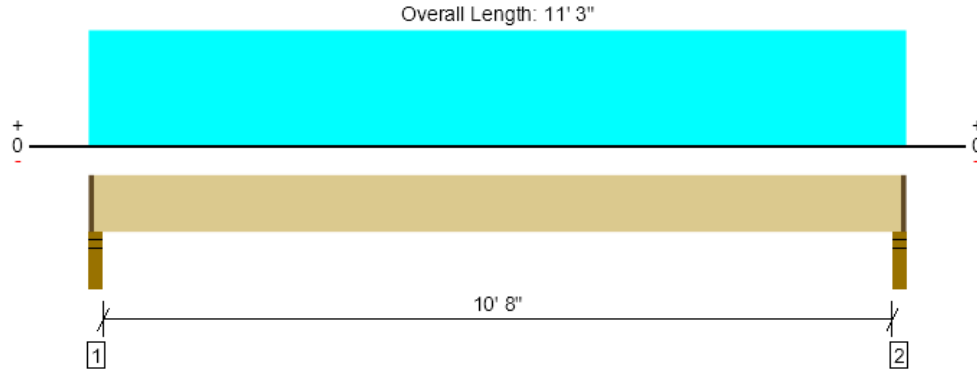
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Level, Deck: Floor Joists

1 piece(s) 2 x 8 DF No.1 @ 16" OC



Drawing is Conceptual. All locations are measured from the outside face of left support (or left cantilever end). All dimensions are horizontal (typ.).

Design Results	Actual @ Location	Allowed	Result	LDF	Load: Combination (Pattern)
Member Reaction (lbs)	442 @ 2 1/2"	1434 (2.25")	Passed (31%)	--	1.0 D + 1.0 L (All Spans)
Shear (lbs)	378 @ 10 3/4"	1305	Passed (29%)	1.00	1.0 D + 1.0 L (All Spans)
Moment (Ft-lbs)	1174 @ 5' 7 1/2"	1511	Passed (78%)	1.00	1.0 D + 1.0 L (All Spans)
Live Load Defl. (in)	0.204 @ 5' 7 1/2"	0.271	Passed (L/637)	--	1.0 D + 1.0 L (All Spans)
Total Load Defl. (in)	0.306 @ 5' 7 1/2"	0.542	Passed (L/425)	--	1.0 D + 1.0 L (All Spans)
TJ-Pro™ Rating	N/A	N/A	N/A	--	N/A

Member Length : 11' 1/2"
System : Floor
Member Type : Joist
Building Use : Residential
Building Code : IBC 2024
Design Methodology : ASD

- Deflection criteria: LL (L/480) and TL (L/240).
- Allowed moment does not reflect the adjustment for the beam stability factor.
- A 15% increase in the moment capacity has been added to account for repetitive member usage.
- Applicable calculations are based on NDS.
- No composite action between deck and joist was considered in analysis.

Supports	Bearing Length			Loads to Supports (lbs)			Accessories
	Total	Available	Required	Dead	Floor Live	Factored	
1 - Stud wall - SPF	3.50"	2.25"	1.50"	150	300	450	1 1/4" Rim Board
2 - Stud wall - SPF	3.50"	2.25"	1.50"	150	300	450	1 1/4" Rim Board

Rim Board is assumed to carry all loads applied directly above it, bypassing the member being designed.

Lateral Bracing	Bracing Intervals	Comments
Top Edge (Lu)	8' 3" o/c	
Bottom Edge (Lu)	11' 1" o/c	

Maximum allowable bracing intervals based on applied load.

Vertical Loads	Location (Side)	Spacing	Dead (0.90)	Floor Live (1.00)	Comments
1 - Uniform (PSF)	0 to 11' 3"	16"	20.0	40.0	Default Load
2 - Uniform (PSF)	0 to 11' 3"	16"	--	--	

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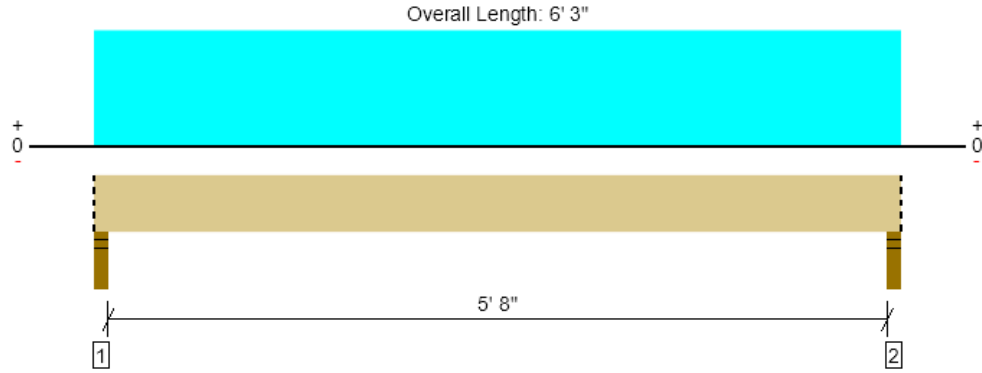
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Level, Deck Beams
1 piece(s) 2 x 8 DF No.1



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Design Results	Actual @ Location	Allowed	Result	LDF	Load: Combination (Pattern)
Member Reaction (lbs)	875 @ 2"	2231 (3.50")	Passed (39%)	--	1.0 D + 1.0 L (All Spans)
Shear (lbs)	624 @ 10 3/4"	1305	Passed (48%)	1.00	1.0 D + 1.0 L (All Spans)
Moment (Ft-lbs)	1226 @ 3' 1 1/2"	1314	Passed (93%)	1.00	1.0 D + 1.0 L (All Spans)
Live Load Defl. (in)	0.073 @ 3' 1 1/2"	0.197	Passed (L/977)	--	1.0 D + 1.0 L (All Spans)
Total Load Defl. (in)	0.095 @ 3' 1 1/2"	0.296	Passed (L/744)	--	1.0 D + 1.0 L (All Spans)

- Deflection criteria: LL (L/360) and TL (L/240).
- Allowed moment does not reflect the adjustment for the beam stability factor.
- Applicable calculations are based on NDS.

Member Length : 6' 3"
System : Floor
Member Type : Drop Beam
Building Use : Residential
Building Code : IBC 2024
Design Methodology : ASD

Supports	Bearing Length			Loads to Supports (lbs)			
	Total	Available	Required	Dead	Floor Live	Factored	Accessories
1 - Stud wall - SPF	3.50"	3.50"	1.50"	209	667	875	Blocking
2 - Stud wall - SPF	3.50"	3.50"	1.50"	209	667	875	Blocking

- Blocking Panels are assumed to carry no loads applied directly above them and the full load is applied to the member being designed.

Lateral Bracing	Bracing Intervals	Comments
Top Edge (Lu)	5' 1" o/c	
Bottom Edge (Lu)	6' 3" o/c	

- Maximum allowable bracing intervals based on applied load.

Vertical Loads	Location (Side)	Tributary Width	Dead (0.90)	Floor Live (1.00)	Comments
0 - Self Weight (PLF)	0 to 6' 3"	N/A	2.8	--	
1 - Uniform (PSF)	0 to 6' 3" (Front)	5' 4"	12.0	40.0	Default Load

- Side loads are assumed to not induce cross-grain tension.

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Level, Deck Posts
1 piece(s) 4 x 4 DF No.1

Post Height: 6'



Design Results	Actual	Allowed	Result	LDF	Load: Combination
Slenderness	21	50	Passed (41%)	--	--
Compression (lbs)	976	11783	Passed (8%)	1.00	1.0 D + 1.0 L
Base Bearing (lbs)	976	363825	Passed (0%)	--	1.0 D + 1.0 L
Bending/Compression	0.06	1	Passed (6%)	1.00	1.0 D + 1.0 L

- Input axial load eccentricity for this design is 16.67% of applicable member side dimension.
- Applicable calculations are based on NDS.

Supports	Type	Material
Base	Plate	Steel

Member Type : Free Standing Post
Building Code : IBC 2024
Design Methodology : ASD

Max Unbraced Length	Comments
Full Member Length	No bracing assumed.

Drawing is Conceptual

Vertical Loads	Dead (0.90)	Floor Live (1.00)	Comments
1 - Point (lb)	20	80	Default Load
2 - Point (lb)	--	--	
3 - Point (lb)	209	667	Linked from: Deck Beams, Support 1

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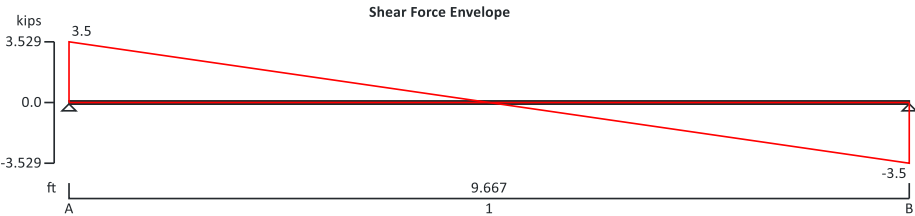
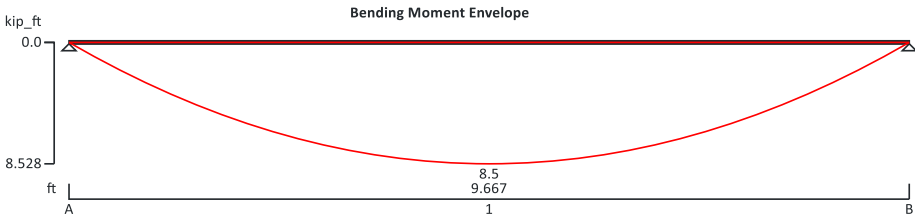
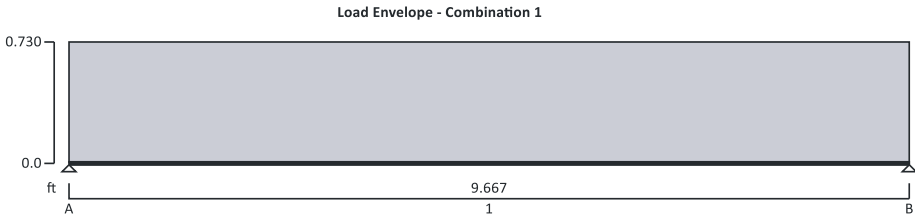
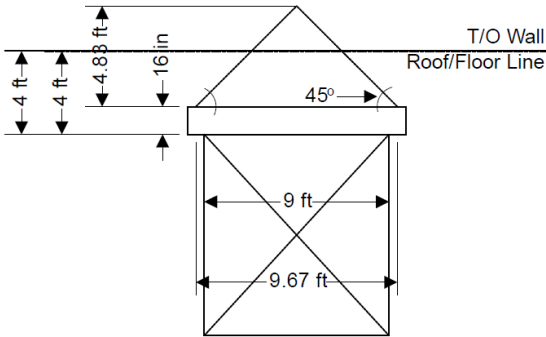
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Support conditions

Support A	Vertically restrained Rotationally free
Support B	Vertically restrained Rotationally free

Applied loading

Beam loads	Dead self weight of beam $\times 1$ Wall - Dead full UDL 147 lb/ft Roof LL - Live full UDL 270 lb/ft Roof DL - Dead full UDL 203 lb/ft
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Load combinations

Load combination 1	Support A	Dead $\times 1.00$ Live $\times 1.00$ Dead $\times 1.00$ Live $\times 1.00$
	Support B	Dead $\times 1.00$ Live $\times 1.00$

Analysis results

Maximum moment;	$M_{\max} = 8.528$ kip_ft;	$M_{\min} = 0$ kip_ft
Maximum shear;	$V_{\max} = 3.53$ kips;	$V_{\min} = -3.53$ kips
Maximum reaction at support A;	$R_{A_{\max}} = 3.5$ kips;	$R_{A_{\min}} = 3.5$ kips
Unfactored dead load reaction at support A;	$R_{A_{\text{Dead}}} = 2.2$ kips	
Unfactored live load reaction at support A;	$R_{A_{\text{Live}}} = 1.3$ kips	
Maximum reaction at support B;	$R_{B_{\max}} = 3.5$ kips;	$R_{B_{\min}} = 3.5$ kips
Unfactored dead load reaction at support B;	$R_{B_{\text{Dead}}} = 2.2$ kips	
Unfactored live load reaction at support B;	$R_{B_{\text{Live}}} = 1.3$ kips	

Masonry details

Masonry type;	Concrete
Density of masonry unit;	$\gamma = 115$ lb/ft ³
Pattern bond;	Running
Mortar type;	PCL Type M
Compressive strength of masonry unit;	$f_{cu} = 1900$ psi
Net compressive strength of masonry (Table 2);	$f_m = 1500$ psi
Modulus of elasticity (1.8.2.2.1);	$E_m = 900 \times f_m = 1350000$ psi
Allowable flexural tensile stress; (Table 2.2.3.2);	$F_t = 106$ psi

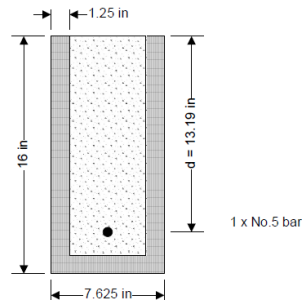
Reinforcement details

Allowable tensile stress;	$F_s = 32000$ psi
Modulus of elasticity of steel;	$E_s = 29000000$ psi

Cover to reinforcement

Bottom cover to reinforcement;	$C_{nom_b} = 1.5$ in
Side cover to reinforcement;	$C_{nom_s} = 1.5$ in

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Section properties

Modular ratio;

$$n = E_s / E_m = 21.48$$

Section width;

$$b = 7.625 \text{ in}$$

Section depth;

$$h = 16 \text{ in}$$

Net shear area;

$$A_{nv} = b \times d = 100.57 \text{ in}^2$$

Section modulus;

$$S = b \times h^2 / 6 = 325.33 \text{ in}^3$$

Depth to tension reinforcement;

$$d = 13.19 \text{ in}$$

Moment of inertia of net section;

$$I_n = b \times h^3 / 12 = 2602.7 \text{ in}^4$$

Cracked moment of inertia;

$$I_{cr} = b \times (k \times d)^3 / 3 + n \times A_s \times (d - k \times d)^2 = 725.1 \text{ in}^4$$

Flexure design (Chapter 2)

Tension reinforcement;

$$1 \times \text{No. 5 bar}$$

Area of tension reinforcement;

$$A_s = N_{bot} \times \text{BarArea}_{bot} = 0.31 \text{ in}^2$$

Reinforcement ratio;

$$\rho_{ratio} = A_s / (b \times d) = 0.00308$$

Neutral axis factor ;

$$k = \sqrt{(2 \times \rho_{ratio} \times n + (\rho_{ratio} \times n)^2)} - \rho_{ratio} \times n = 0.304$$

Lever arm factor;

$$j = 1 - k / 3 = 0.899$$

Cracking moment;

$$M_{cr} = 2.5 \times F_t \times S = 7.2 \text{ kip_ft}$$

Design bending moment;

$$M = 8.53 \text{ kip_ft}$$

Tensile stress in reinforcement;

$$f_s = M / (A_s \times j \times d) = 27848 \text{ psi}$$

Allowable tensile stress in reinf. (2.3.3.1);

$$F_s = 32000 \text{ psi}$$

$$f_s / F_s = 0.870$$

PASS - Allowable tensile stress exceeds tensile stress due to flexure

Compressive stress in masonry;

$$f_b = 2 \times M / (j \times k \times b \times d^2) = 565.3 \text{ psi}$$

Allowable stress in masonry (2.3.4.2.2);

$$F_b = 0.45 \times f'_m = 675.0 \text{ psi}$$

$$f_b / F_b = 0.838$$

PASS - Allowable compressive stress exceeds compressive stress due to flexure

Shear design (Chapter 2)

Location of shear force;

Face of support

Design shear force;

$$V = 3.29 \text{ kips}$$

Depth of shear area;

$$d = 13.19 \text{ in}$$

Moment shear relationship, M/Vd;

$$\text{Assume; } M_Vd_{ratio} = 1$$

Shear stress (2-24);

$$f_v = V / A_{nv} = 32.7 \text{ psi}$$

Allowable masonry shear stress (2-28);

$$F_v = 1/2 \times (4.0 - 1.75 \times M_Vd_{ratio}) \times \sqrt{f'_m \times 1\text{psi}}$$

$$F_v = \mathbf{43.6 \text{ psi}}$$

Masonry shear stress ratio;

$$f_v / F_v = \mathbf{0.750}$$

PASS - Allowable shear stress exceeds shear stress in masonry
Deflection (1.13.1.4)

Deflection load combination;

Dead + Live

Moment at midspan under deflection loads;

$$M_a = \mathbf{8.528 \text{ kip_ft}}$$

Effective moment of inertia (1-1);

$$I_{eff} = \min(I_n \times (M_{cr} / M_a)^3 + I_{cr} \times (1 - (M_{cr} / M_a)^3), I_n) = \mathbf{1847.5 \text{ in}^4}$$

Maximum instantaneous deflection;

$$\delta_{inst} = \mathbf{0.06 \text{ in; or span / 2017}}$$

Deflection limit;

$$\delta_{limit} = \text{span} / 600$$

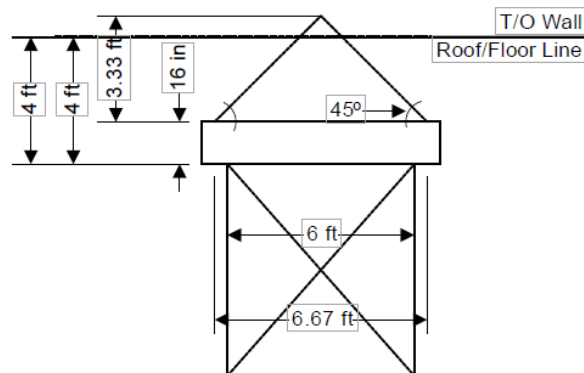
PASS - Deflection limit exceeds deflection of lintel

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MASONRY LINTEL ANALYSIS AND DESIGN TO TMS/MSJC 2011

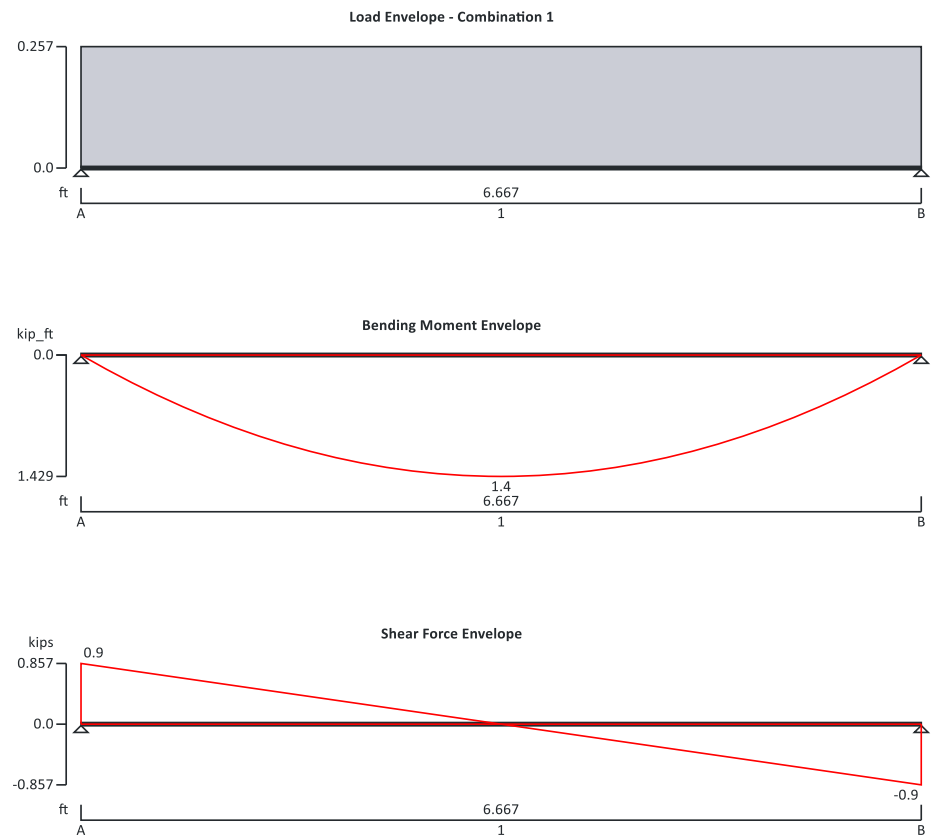
Tedds calculation version 1.2.05

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Support conditions

Support A	Vertically restrained Rotationally free
Support B	Vertically restrained Rotationally free

Applied loading

Beam loads	Dead self weight of beam × 1 Wall - Dead full UDL 147 lb/ft
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Load combinations

Load combination 1	Support A	Dead × 1.00
		Live × 1.00
		Dead × 1.00
	Support B	Live × 1.00
		Dead × 1.00
		Live × 1.00

Analysis results

Maximum moment;	$M_{\max} = 1.429 \text{ kip_ft};$	$M_{\min} = 0 \text{ kip_ft}$
Maximum shear;	$V_{\max} = 0.86 \text{ kips};$	$V_{\min} = -0.86 \text{ kips}$

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Maximum reaction at support A;	$R_{A_max} = 0.9$ kips;	$R_{A_min} = 0.9$ kips
Unfactored dead load reaction at support A;	$R_{A_Dead} = 0.9$ kips	
Maximum reaction at support B;	$R_{B_max} = 0.9$ kips;	$R_{B_min} = 0.9$ kips
Unfactored dead load reaction at support B;	$R_{B_Dead} = 0.9$ kips	

Masonry details

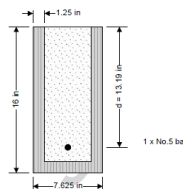
Masonry type;	Concrete
Density of masonry unit;	$\gamma = 115$ lb/ft ³
Pattern bond;	Running
Mortar type;	PCL Type M
Compressive strength of masonry unit;	$f_{cu} = 1900$ psi
Net compressive strength of masonry (Table 2);	$f_m = 1500$ psi
Modulus of elasticity (1.8.2.2.1);	$E_m = 900 \times f_m = 1350000$ psi
Allowable flexural tensile stress; (Table 2.2.3.2);	$F_t = 106$ psi

Reinforcement details

Allowable tensile stress;	$F_s = 32000$ psi
Modulus of elasticity of steel;	$E_s = 29000000$ psi

Cover to reinforcement

Bottom cover to reinforcement;	$C_{nom_b} = 1.5$ in
Side cover to reinforcement;	$C_{nom_s} = 1.5$ in



Section properties

Modular ratio;	$n = E_s / E_m = 21.48$
Section width;	$b = 7.625$ in
Section depth;	$h = 16$ in
Net shear area;	$A_{nv} = b \times d = 100.57$ in ²
Section modulus;	$S = b \times h^2 / 6 = 325.33$ in ³
Depth to tension reinforcement;	$d = 13.19$ in
Moment of inertia of net section;	$I_n = b \times h^3 / 12 = 2602.7$ in ⁴
Cracked moment of inertia;	$I_{cr} = b \times (k \times d)^3 / 3 + n \times A_s \times (d - k \times d)^2 = 725.1$ in ⁴

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Flexure design (Chapter 2)

Tension reinforcement;	1 x No. 5 bar
Area of tension reinforcement;	$A_s = N_{bot} \times BarArea_{bot} = 0.31 \text{ in}^2$
Reinforcement ratio;	$\rho_{ratio} = A_s / (b \times d) = 0.00308$
Neutral axis factor ;	$k = \sqrt{(2 \times \rho_{ratio} \times n + (\rho_{ratio} \times n)^2)} - \rho_{ratio} \times n = 0.304$
Lever arm factor;	$j = 1 - k / 3 = 0.899$
Cracking moment;	$M_{cr} = 2.5 \times F_t \times S = 7.2 \text{ kip_ft}$
Design bending moment;	$M = 1.43 \text{ kip_ft}$
Tensile stress in reinforcement;	$f_s = M / (A_s \times j \times d) = 4665 \text{ psi}$
Allowable tensile stress in reinf. (2.3.3.1);	$F_s = 32000 \text{ psi}$ $f_s / F_s = 0.146$
	PASS - Allowable tensile stress exceeds tensile stress due to flexure
Compressive stress in masonry;	$f_b = 2 \times M / (j \times k \times b \times d^2) = 94.7 \text{ psi}$
Allowable stress in masonry (2.3.4.2.2);	$F_b = 0.45 \times f'_m = 675.0 \text{ psi}$ $f_b / F_b = 0.140$

PASS - Allowable compressive stress exceeds compressive stress due to flexure

Shear design (Chapter 2)

Location of shear force;	Face of support
Design shear force;	$V = 0.77 \text{ kips}$
Depth of shear area;	$d = 13.19 \text{ in}$
Moment shear relationship, M/Vd;	Assume; $M_Vd_{ratio} = 1$
Shear stress (2-24);	$f_v = V / A_{nv} = 7.7 \text{ psi}$
Allowable masonry shear stress (2-28);	$F_v = 1/2 \times (4.0 - 1.75 \times M_Vd_{ratio}) \times \sqrt{f'_m \times 1 \text{ psi}}$ $F_v = 43.6 \text{ psi}$
Masonry shear stress ratio;	$f_v / F_v = 0.176$
	PASS - Allowable shear stress exceeds shear stress in masonry

Deflection (1.13.1.4)

Deflection load combination;	Dead
Moment at midspan under deflection loads;	$M_a = 1.429 \text{ kip_ft}$
Effective moment of inertia (1-1);	$I_{eff} = \min(I_n \times (M_{cr} / M_a)^3 + I_{cr} \times (1 - (M_{cr} / M_a)^3), I_n) = 2602.7 \text{ in}^4$
Maximum instantaneous deflection;	$\delta_{inst} = 0 \text{ in}; \text{ or } span / 24596$
Deflection limit;	$\delta_{limit} = span / 600$

PASS - Deflection limit exceeds deflection of lintel

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RC BEAM DESIGN (ACI318-2019 (22))

In accordance with ACI318-2019 (22)

Tedds calculation version 3.3.23

Concrete details

Compressive strength of concrete;	$f'_c = 3000 \text{ psi}$
Density of reinforced concrete;	$w_c = 150 \text{ lb / ft}^3$
Concrete type;	Normal weight
Modulus of elasticity of concrete (cl.19.2.2.1);	$E = (w_c / 1 \text{ lb/ft}^3)^{1.5} \times 33 \text{ psi} \times (f'_c / 1 \text{ psi})^{0.5} = 3320561 \text{ psi}$

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Strength reduction factor for shear;

$$\phi_s = 0.75$$

Reinforcement details

Yield strength of reinforcement;

$$f_y = 60000 \text{ psi}$$

Compression-controlled strain limit (cl.21.2.2.1);

$$\epsilon_{ty} = 0.00200$$

Nominal cover to reinforcement

Cover to top reinforcement;

$$c_{nom_t} = 1.5 \text{ in}$$

Cover to bottom reinforcement;

$$c_{nom_b} = 1.5 \text{ in}$$

Cover to side reinforcement;

$$c_{nom_s} = 1.5 \text{ in}$$

Section 1 - Porch Beams

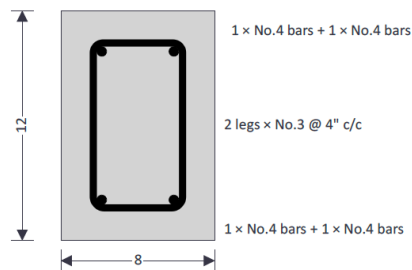
Rectangular section details

Section width;

$$b = 8 \text{ in}$$

Section depth;

$$h = 12 \text{ in}$$



Positive moment. Rectangular section in flexure (Section 9.5.2)

Factored bending moment at section;

$$M_u = M_{pos_s1} = 12.200 \text{ kip_ft}$$

Effective depth to tension reinforcement;

$$d = 9.875 \text{ in}$$

Tension reinforcement provided;

$$1 \times \text{No.4 bars} + 1 \times \text{No.4 bars}$$

Area of tension reinforcement provided;

$$A_{s,prov} = 0.393 \text{ in}^2$$

Minimum area of reinforcement (cl.9.6.1.2);

$$A_{s,min} = \max(3 \text{ psi} \times \sqrt{f'_c / 1 \text{ psi}}, 200 \text{ psi}) \times b \times d / f_y = 0.263 \text{ in}^2$$

PASS - Area of reinforcement provided is greater than minimum area of reinforcement required

Stress block depth factor (cl.22.2.2.4.3);

$$\beta_1 = \min(\max(0.85 - 0.05 \times (f'_c - 4 \text{ ksi}) / 1 \text{ ksi}, 0.65), 0.85) = 0.85$$

Depth of equivalent rectangular stress block;

$$a = A_{s,prov} \times f_y / (0.85 \times f'_c \times b) = 1.155 \text{ in}$$

Depth to neutral axis;

$$c = a / \beta_1 = 1.359 \text{ in}$$

Net tensile strain in extreme tension fibers;

$$\epsilon_t = 0.003 \times (d_o - c) / \max(c, 0.001 \text{ in}) = 0.0188$$

Net tensile strain in tension controlled zone

Strength reduction factor (cl.21.2.1);

$$\phi_f = \min(\max(0.65 + 0.25 \times (\epsilon_t - \epsilon_{ty}) / (0.003), 0.65), 0.9) = 0.90$$

Nominal moment strength;

$$M_n = A_{s,prov} \times f_y \times (d - a / 2) = 18.256 \text{ kip_ft}$$

Design moment strength;

$$\phi M_n = M_n \times \phi_f = 16.430 \text{ kip_ft}$$

PASS - Required moment strength is less than design moment strength

Flexural cracking

Max. center to center spacing of tension reinf.;

$$s_{b,max} = s_{bot} + \phi_{s1_add_b_L1} = 3.750 \text{ in}$$

Service load stress in reinforcement (cl.24.3.2);

$$f_s = 2/3 \times f_y = 40000 \text{ psi}$$

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Clear cover of reinforcement; $C_c = C_{nom_b} + \phi_v = \mathbf{1.875}$ in
Maximum allowable bot bar spacing (Table 24.3.2); $s_{max} = \min(15in \times 40000psi / f_s - 2.5 \times c_c, 12in \times 40000psi / f_s) = \mathbf{10.313}$ in
PASS - Maximum allowable tension reinforcement spacing exceeds actual spacing

Spacing limits for reinforcement

Top bar clear spacing; $s_{top} = (b - (2 \times (C_{nom_s} + \phi_{s1_v}) + \phi_{s1_t_L1} \times N_{s1_t_L1} + \phi_{s1_add_t_L1} \times N_{s1_add_t_L1})) / (N_{s1_t_L1} + N_{s1_add_t_L1} - 1) = \mathbf{3.250}$ in
Min. allowable top bar clear spacing (cl.25.2.1); $s_{top,min} = \mathbf{1.000}$ in
Bottom bar clear spacing; $s_{bot} = (b - (2 \times (C_{nom_s} + \phi_{s1_v}) + \phi_{s1_b_L1} \times N_{s1_b_L1} + \phi_{s1_add_b_L1} \times N_{s1_add_b_L1})) / (N_{s1_b_L1} + N_{s1_add_b_L1} - 1) = \mathbf{3.250}$ in
Min. allowable bottom bar clear spacing (cl.25.2.1); $s_{bot,min} = \mathbf{1.000}$ in
PASS - Actual bar spacing exceeds minimum allowable

Rectangular section in shear

Design shear force; $V_u = \mathbf{4.050}$ kips
Concrete weight modification factor; $\lambda = \mathbf{1.00}$
Size effect factor (cl. 22.5.5.1.3); $\lambda_s = \min(\sqrt{2 / (1 + (d / 1 \text{ in}) / 10)}, 1.0) = \mathbf{1.00}$
Area of longitudinal reinforcement; $A_{st} = \mathbf{0.393}$ in²
Ratio of longitudinal reinforcement; $\rho_w = A_{st} / (b \times d) = \mathbf{0.005}$
Concrete shear strength (Table 22.5.5.1); $\phi V_c = \phi_s \times \min(8 \times \lambda \times (\rho_w)^{1/3}, 5 \times \lambda) \times \sqrt{f'_c \times 1 \text{ psi}} \times b \times d = \mathbf{4.431}$ kips
Reinforcement shear strength (eqn. 22.5.1.1); $\phi V_s = \max(V_u - \phi V_c, 0 \text{ kips}) = \mathbf{0.000}$ kips
Maximum reinforcement shear strength; $\phi V_{s,max} = \phi_s \times 8 \text{ psi} \times \sqrt{(\min(f'_c, 10000psi) / 1 \text{ psi})} \times b \times d = \mathbf{25.962}$ kips
No reinforcement shear strength limit; $\phi V_{s,lim} = \phi_s \times \lambda \times 1 \text{ psi} \times \sqrt{(\min(f'_c, 10000psi) / 1 \text{ psi})} \times b \times d = \mathbf{3.245}$ kips
Area of design shear reinf. reqd (eqn. 22.5.8.5.3); $A_{sv,des} = \phi V_s / (\phi_s \times \min(f_y, 60000 \text{ psi}) \times d) = \mathbf{0.000}$ in²/ft
Min area of shear reinforcement (Table 9.6.3.4); $A_{sv,min} = \max(50 \text{ psi}, 0.75 \text{ psi} \times \sqrt{f'_c / 1 \text{ psi}}) \times b / \min(f_y, 60000 \text{ psi}) = \mathbf{0.080}$ in²/ft
 $\phi V_c \geq V_u \geq \phi V_{s,lim}$ - minimum reinforcement required
Area of shear reinforcement required; $A_{sv,req} = A_{sv,min} = \mathbf{0.080}$ in²/ft
Shear reinforcement provided; $2 \text{ legs} \times \text{No.3 @ } 4" \text{ c/c}$
Area of shear reinforcement provided; $A_{sv,prov} = \mathbf{0.663}$ in²/ft

PASS - Area of shear reinforcement provided exceeds area of shear reinforcement required

Maximum longitudinal spacing (Table 9.7.6.2.2); $s_{vl,max} = \min(d / 2, 24 \text{ in}) = \mathbf{4.937}$ in
PASS - longitudinal spacing of stirrups is less than the maximum allowable

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Seismic properties

Seismic design category; A

Seismic importance factor (ASCE7 Table 1.5-2); $I_e = 1$

Design spectral response acceleration parameter, short periods (IBC 1613.2.4 or ASCE7 11.4.5)

$S_{DS} = 1$

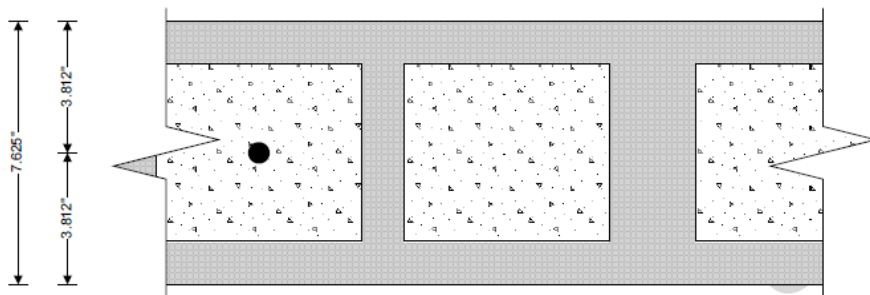
Seismic wall classification; Nonparticipating

No prescriptive minimum seismic reinforcement

Redundancy factor, on out-of-plane load; $\rho_E = 1.0$

Construction details

Wall thickness; $t = 7.625$ in



Masonry details

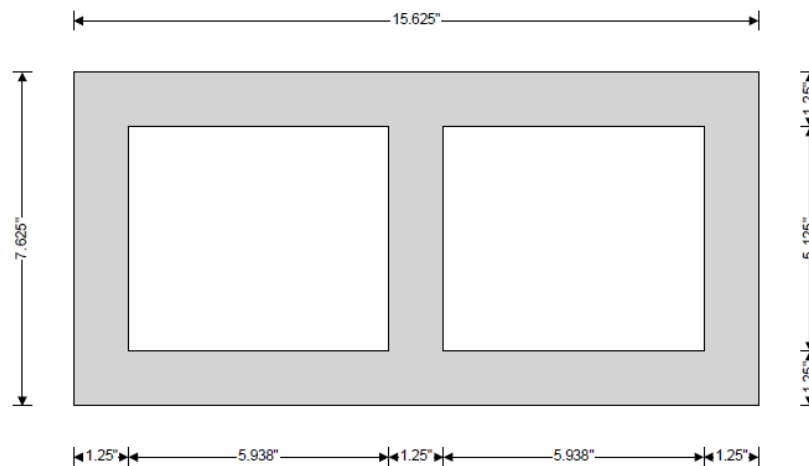
Hollow concrete units fully grouted in running bond fully bedded with PCL class M mortar

Compressive strength of unit; $f_{cu} = 1900$ psi

Density of masonry units; $\gamma_{block} = 115$ lb/ft³

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Height of masonry units;	$h_b = 7.625$ in
Length of masonry units;	$l_b = 15.625$ in
Number of internal webs;	$N_{web} = 1$
Number of end webs;	$N_{end} = 2$
Internal web thickness;	$t_{bw} = 1.25$ in
Face shell thickness;	$t_{bf} = 1.25$ in
End web thickness;	$t_{be} = 1.25$ in
Area of block;	$A_{block} = [t \times l_b - (l_b - N_{web} \times t_{bw} - N_{end} \times t_{be}) \times (t - 2 \times t_{bf})] / l_b = 44.76$ in ² /ft
Area of grout;	$A_{grout} = [(l_b - N_{web} \times t_{bw} - N_{end} \times t_{be}) \times (t - 2 \times t_{bf})] / l_b = 46.74$ in ² /ft
Density of grout;	$\gamma_{grout} = 140$ lb/ft ³
Self weight of wall;	$WSW = A_{block} \times \gamma_{block} + A_{grout} \times \gamma_{grout} = 81.19$ psf



From TMS 602-16 Table 2 - Compressive strength of masonry

Net compressive strength of masonry;	$f'_m = 1900$ psi
Modulus of elasticity for masonry;	$E_m = 900 \times f'_m = 1710000$ psi
Shear modulus of masonry;	$G_v = 0.4 \times E_m = 684000$ psi

From TMS 402 -16 Table 9.1.9.2 - Modulus of rupture

Modulus of rupture normal to bed;	$f_{r_norm} = 163$ psi
Modulus of rupture parallel to bed;	$f_{r_para} = 267$ psi

Reinforcement details

Yield strength of reinforcement;	$f_y = 60000$ psi
Allowable tensile stress in reinforcement;	$F_s = 32000$ psi
Modulus of elasticity for reinforcement;	$E_s = 29000000$ psi
Vertical reinforcement provided;	No.5 bars at 24 in centers
Area of vertical reinforcement;	$A_s = \pi \times Dia^2 / (4 \times s) = 0.15$ in ² /ft

Lateral out-of-plane loads

Wind load on panel;	$W = 40$ psf
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Wind load on parapet; $W_p = 18$ psf
Seismic load factor (ASCE7 12.11.1); $F_p = 0.4 \times S_{DS} \times I_e = 0.4$
Seismic load from wall; $E_{wall} = \max(F_p, 0.1) \times w_{SW} = 32.5$ psf
Additional seismic load; $E_{add} = 0$ psf
Seismic lateral load on panel; $E = E_{wall} + E_{add} = 32.5$ psf

Lateral in-plane loads

Vertical loading details

Vertical seismic load factor applied to dead load; $F_{Ev} = 0.2 \times S_{DS} = 0.2$

From IBC 2021 cl.1605.2 - Basic load combinations (Utilization)

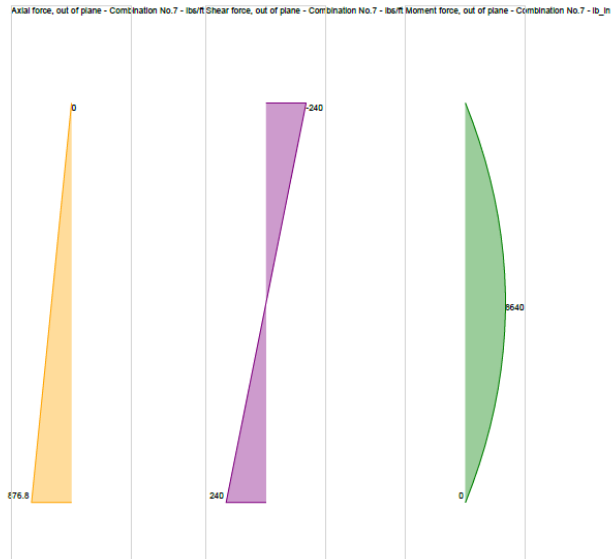
Load combination no.1; $1.4 \times DL$; (0.017)
Load combination no.5; $1.2 \times DL + W + LL + 0.5 \times (LL_r \text{ or } SL \text{ or } RL)$; (0.276)
Load combination no.7; $0.9 \times DL + W$; (0.280)

Properties of masonry section

Cross-sectional area; $A = t = 91.5$ in²/ft
Properties for walls loaded out-of-plane:
Moment of inertia; $I = t^3 / 12 = 443.3$ in⁴/ft
Section modulus; $S = I / c = 116.3$ in³/ft
Radius of gyration; $r = \sqrt{I / A} = 2.201$ in
Effective height factor; $K = 1$

Consider wall at maximum moment location under load combination no.7

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Maximum moment location;	6 ft
Axial load at mid-height of panel;	$P = 438 \text{ lb/ft}$
Slenderness ratio;	$(K \times h) / r = 65.420; < 99$
Nominal axial strength;	$P_n = 0.8 \times (0.8 \times f_m \times (A - A_s) + A_s \times 0 \text{ ksi}) \times [1 - ((K \times h) / (140 \times r))^2] = 86823 \text{ lb/ft}$
Strength reduction factor;	$\phi = 0.9$
Design axial strength;	$\phi \times P_n = 78140 \text{ lb/ft}$ $P / (\phi \times P_n) = 0.006$
	PASS - Nominal axial strength exceeds axial load
Factored axial stress;	$P / A = 5 \text{ psi}$
Factored axial stress limit;	$0.2 \times f_m = 380 \text{ psi}$
	PASS - Allowable stress under out of plane loads exceeds factored axial stress
Nominal cracking moment strength;	$M_{cr} = S \times f_{r_norm} = 18954 \text{ lb_in/ft}$
Modular ratio;	$n = E_s / E_m = 16.959$
Distance to neutral axis;	$c_{cr} = (A_s \times f_y + P) / (0.64 \times f_m) = 0.661 \text{ in}$
Moment of inertia of cracked section;	$I_{cr} = n \times (A_s \times P \times t / (f_y \times 2 \times d)) \times (d - c_{cr})^2 + c_{cr}^3 / 3 = 28.2 \text{ in}^4/\text{ft}$
By iteration;	$M_{u0} = M = 8640 \text{ lb_in/ft}$ $\delta_{u0} = 5 \times M_{u0} \times h^2 / (48 \times E_m \times I) = 0.025 \text{ in}$ $M_{u1} = M_{u0} + P \times \delta_{u0} = 8651 \text{ lb_in/ft}$ $\delta_{u1} = 5 \times M_{u1} \times h^2 / (48 \times E_m \times I) = 0.025 \text{ in}$
;	

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Bending moment at mid-height of panel;

$$M = M_{u0} + P \times \delta_{u1} = \mathbf{8651 \text{ lb_in/ft}}$$

Depth of reinforcement;

$$d = \mathbf{3.813 \text{ in}}$$

Effective width per bar;

$$b_{\text{eff}} = \min(s, 6 \times t_{\text{nom}}, 72 \text{ in}) = \mathbf{24 \text{ in}}$$

Tensile strain in reinforcement;

$$\epsilon_s = f_y / E_s = \mathbf{0.0021}$$

Maximum usable compressive strain of masonry;

$$\epsilon_{mu} = \mathbf{0.0025}$$

Fiber of max.compressive strain to neutral axis;

$$C_{\text{bal}} = \epsilon_{mu} \times d / (\epsilon_{mu} + \epsilon_s) = \mathbf{2.086 \text{ in}}$$

Net tensile strain;

$$\epsilon_t = 0.003 \times (d - C_{\text{bal}}) / C_{\text{bal}} = \mathbf{0.0025}$$

Strength reduction factor;

$$\phi = \mathbf{0.9}$$

Tensile force at balance point;

$$T_{\text{bal}} = A_s \times f_y = \mathbf{9204 \text{ lb/ft}}$$

$$\beta_1 = \mathbf{0.8}$$

Compressive force at balance point;

$$C_{\text{bal}} = 0.8 \times f'_m \times \beta_1 \times C_{\text{bal}} = \mathbf{30440 \text{ lb/ft}}$$

Design axial force at balance point;

$$P_{\text{bal}} = \phi \times (C_{\text{bal}} - T_{\text{bal}}) = \mathbf{19113 \text{ lb/ft}}$$

Design moment at balance point;

$$M_{\text{bal}} = \phi \times [T_{\text{bal}} \times (d - t / 2) + C_{\text{bal}} \times (t / 2 - \beta_1 \times C_{\text{bal}} / 2)] = \mathbf{81588 \text{ lb_in/ft}}$$

Maximum design moment from interaction diagram;

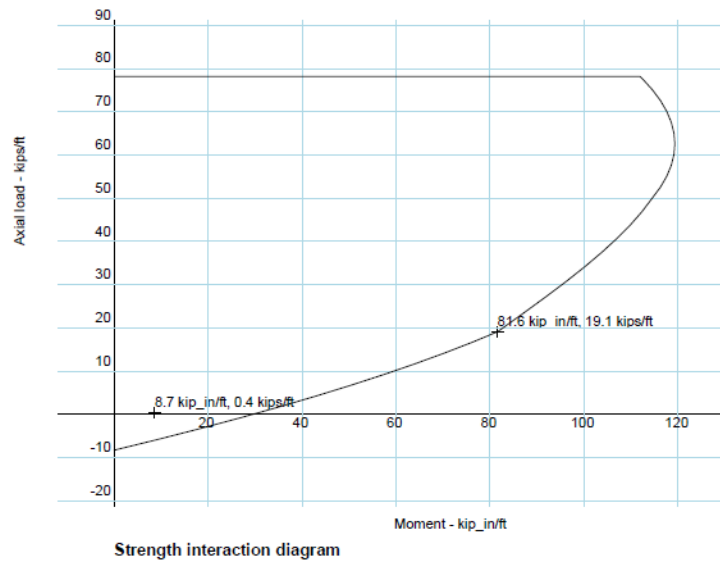
$$M_c = \mathbf{30888 \text{ lb_in/ft}}$$

$$M / M_c = \mathbf{0.280}$$

PASS - Combination of applied axial load and flexure is acceptable

Maximum area of tensile reinforcement (9.3.3.2);

$$A_{s_{\text{max}}} = 0.64 \times f'_m \times [\epsilon_{mu} / (\epsilon_{mu} + 1.5 \times \epsilon_s)] \times d / f_y = \mathbf{0.414 \text{ in}^2/\text{ft}}$$

PASS - Area of reinforcement provided is less than maximum allowable

Consider wall at top under load combination no.7

Shear force;

$$V = \mathbf{240 \text{ lb/ft}}$$

Compressive force;

$$N_u = \mathbf{0 \text{ lb/ft}}$$

Net shear area;

$$A_{nv} = d = \mathbf{45.750 \text{ in}^2/\text{ft}}$$

Nominal shear strength;

$$V_n = \min([4 - 1.75 \times \min(M / (V \times d), 1)] \times A_{nv} \times \sqrt{(f'_m \times 1 \text{ psi})} + 0.25 \times N_u, 6 \times A_{nv} \times \sqrt{(f'_m \times 1 \text{ psi})}) = \mathbf{7977 \text{ lb/ft}}$$

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FOOTING ANALYSIS

In accordance with ACI318-19 (22)

Tedds calculation version 3.3.14

Summary results

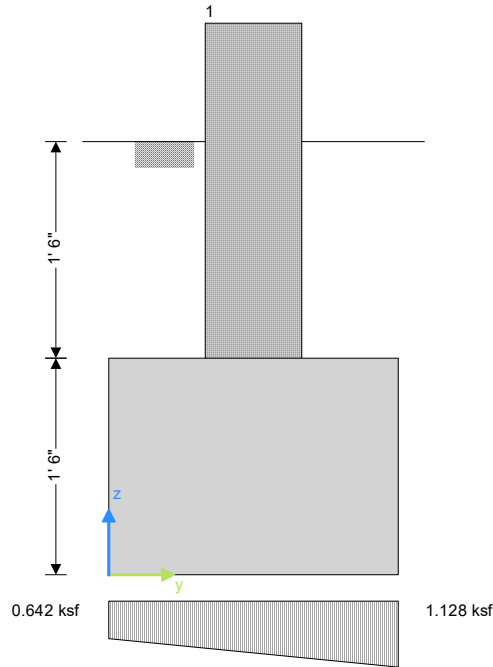
Overall design status **PASS**
Overall design utilization **0.752**

Description	Unit	Applied	Resisting	FoS	Result
Uplift verification	kips	1.8			Pass
Overturning stability, y	kip_ft	0.16	-1.77	10.93	Pass
Sliding stability, y	kips	0.1	1.0	9.462	Pass
Description	Unit	Applied	Resisting	Utilization	Result
Soil bearing	ksf	1.128	1.5	0.752	Pass
Description	Unit	Required	Provided	Utilization	Result
Moment, positive, y-direction	kip_ft	0.1	30.0	0.004	Pass
Min.area of reinf, bot., y-direction	in ²	0.389	0.465		Pass
Max.reinf.spacing, bot, y-direction	in	18.0	8.0		Pass
Min.area of reinf, S&T, x-direction	in ²	0.778	0.930		Pass
Max.reinf.spacing, S&T, x-direction	in	18.0	8.0		Pass

Strip footing details - considering a one foot strip

Length of footing $L_x = 1$ ft
Width of footing $L_y = 2$ ft
Footing area $A = L_x \times L_y = 2$ ft²
Depth of footing $h = 18$ in
Depth of soil over footing $h_{soil} = 18$ in
Density of concrete $\gamma_{conc} = 150.0$ lb/ft³

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Wall no.1 details

Width of wall
 position in y-axis

Soil properties

Gross allowable bearing pressure
 Density of soil
 Angle of internal friction
 Design base friction angle
 Coefficient of base friction
 Self weight
 Soil weight

Wall no.1 loads per linear foot

Dead load in z
 Live roof load in z
 Wind load in y

Footing analysis for soil and stability

Load combinations per ASCE 7-16

1.0D (0.520)
 1.0D + 1.0L (0.520)
 1.0D + 1.0Lr (0.613)
 1.0D + 0.6W (0.736)
 1.0D + 0.75L + 0.75Lr + 0.45W (0.752)
 0.6D + 0.6W (0.528)

$l_{y1} = 8 \text{ in}$
 $y_1 = 12 \text{ in}$

$Q_{allow_Gross} = 1.5 \text{ ksf}$
 $\gamma_{soil} = 120.0 \text{ lb/ft}^3$
 $\phi_b = 30.0 \text{ deg}$
 $\delta_{bb} = 30.0 \text{ deg}$
 $\tan(\delta_{bb}) = 0.577$
 $F_{swt} = h \times \gamma_{conc} = 225 \text{ psf}$
 $F_{soil} = h_{soil} \times \gamma_{soil} = 180 \text{ psf}$

 $F_{Dz1} = 0.9 \text{ kips}$
 $F_{Lr,z1} = 0.3 \text{ kips}$
 $F_{Wy1} = 0.2 \text{ kips}$

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Combination 11 results: 1.0D + 0.75L + 0.75Lr + 0.45W

Forces on footing per linear foot

Force in y-axis

Force in z-axis

$$F_{dy} = \gamma_w \times F_{wy1} = \mathbf{0.1 \text{ kips}}$$

$$F_{dz} = \gamma_D \times A \times (F_{swt} + F_{soil}) + \gamma_D \times (F_{Dz1} - l_{x1} \times l_{y1} \times h_{soil} \gamma_{soil}) + \gamma_{Lr} \times F_{Lrz1} = \mathbf{1.8 \text{ kips}}$$

Moments on footing per linear foot

Moment in y-axis, about y is 0

$$M_{dy} = \gamma_D \times (A \times (F_{swt} + F_{soil}) \times L_y / 2) + \gamma_D \times (((F_{Dz1} - l_{x1} \times l_{y1} \times h_{soil} \gamma_{soil})) \times y_1) + \gamma_{Lr} \times (F_{Lrz1} \times y_1) + \gamma_w \times (F_{wy1} \times (h)) = \mathbf{1.9 \text{ kip_ft}}$$

Uplift verification

Vertical force

$$F_{dz} = \mathbf{1.77 \text{ kips}}$$

Overturning moment

Resisting moment

PASS - Footing is not subject to uplift

Stability against overturning in y direction, moment about y is L_y

$$M_{OTyL} = \gamma_w \times (F_{wy1} \times (h)) = \mathbf{0.16 \text{ kip_ft}}$$

Factor of safety

$$M_{RyL} = -1 \gamma_D \times (A \times (F_{swt} + F_{soil}) \times L_y / 2) + \gamma_D \times (((F_{Dz1} - l_{x1} \times l_{y1} \times h_{soil} \gamma_{soil})) \times (y_1 - L_y)) + \gamma_{Lr} \times (F_{Lrz1} \times (y_1 - L_y)) = \mathbf{-1.77 \text{ kip_ft}}$$

Stability against sliding

Resistance due to base friction

$$\text{abs}(M_{RyL} / M_{OTyL}) = \mathbf{10.926}$$

PASS - Overturning moment safety factor exceeds the minimum of 1.00

Stability against sliding in y direction

Total sliding resistance

Factor of safety

$$F_{R\text{Friction}} = \max(F_{dz}, 0 \text{ kN}) \times \tan(\delta_{bb}) = \mathbf{1.022 \text{ kips}}$$

$$F_{Ry} = F_{R\text{Friction}} = \mathbf{1.022 \text{ kips}}$$

$$\text{abs}(F_{Ry} / F_{dy}) = \mathbf{9.46}$$

Bearing resistance

Eccentricity of base reaction

Eccentricity of base reaction in y-axis

PASS - Sliding factor of safety exceeds the minimum of 1.00

Strip base pressures

$$e_{dy} = M_{dy} / F_{dz} - L_y / 2 = \mathbf{1.098 \text{ in}}$$

Minimum base pressure

Maximum base pressure

$$q_1 = F_{dz} \times (1 - 6 \times e_{dy} / L_y) / (L_y \times 1 \text{ ft}) = \mathbf{0.642 \text{ ksf}}$$

$$q_2 = F_{dz} \times (1 + 6 \times e_{dy} / L_y) / (L_y \times 1 \text{ ft}) = \mathbf{1.128 \text{ ksf}}$$

$$q_{\min} = \min(q_1, q_2) = \mathbf{0.642 \text{ ksf}}$$

$$q_{\max} = \max(q_1, q_2) = \mathbf{1.128 \text{ ksf}}$$

Allowable bearing capacity

Allowable bearing capacity

$$q_{\text{allow}} = q_{\text{allow_Gross}} = \mathbf{1.5 \text{ ksf}}$$

$$q_{\max} / q_{\text{allow}} = \mathbf{0.752}$$

PASS - Allowable bearing capacity exceeds design base pressure

FOOTING DESIGN

In accordance with ACI318-19 (22)

Material details

Compressive strength of concrete

Yield strength of reinforcement

$$f_c = \mathbf{4000 \text{ psi}}$$

$$f_y = \mathbf{60000 \text{ psi}}$$

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Compression-controlled strain limit (21.2.2)

$$\epsilon_{ty} = 0.00200$$

Cover to top of footing

$$C_{nom_t} = 3 \text{ in}$$

Cover to side of footing

$$C_{nom_s} = 3 \text{ in}$$

Cover to bottom of footing

$$C_{nom_b} = 3 \text{ in}$$

Concrete type

Normal weight

Concrete modification factor

$$\lambda = 1.00$$

Wall type

Concrete

Analysis and design of concrete footing

Load combinations per ASCE 7-16

1.4D (0.004)

$$F_{uz} = \gamma_D \times A \times (F_{swt} + F_{soil}) + \gamma_D \times (F_{Dz1} - l_{x1} \times l_{y1} \times h_{soil} \times \gamma_{soil}) = 2.2 \text{ kips}$$

1.2D + 1.6L + 0.5Lr (0.004)

Combination 1 results: 1.4D

$$M_{uy} = \gamma_D \times (A \times (F_{swt} + F_{soil}) \times L_y / 2) + \gamma_D \times (((F_{Dz1} - l_{x1} \times l_{y1} \times h_{soil} \times \gamma_{soil})) \times y_1) = 2.2 \text{ kip_ft}$$

Forces on footing per linear foot

Ultimate force in z-axis

$$e_{uy} = M_{uy} / F_{uz} - L_y / 2 = 0.000 \text{ in}$$

Moments on footing per linear foot

Ultimate moment in y-axis, about y is 0

$$q_{u1} = F_{uz} \times (1 - 6 \times e_{uy} / L_y) / (L_y \times 1 \text{ ft}) = 1.092 \text{ ksf}$$

$$q_{u2} = F_{uz} \times (1 + 6 \times e_{uy} / L_y) / (L_y \times 1 \text{ ft}) = 1.092 \text{ ksf}$$

Eccentricity of base reaction

Eccentricity of base reaction in y-axis

$$q_{umin} = \min(q_{u1}, q_{u2}) = 1.092 \text{ ksf}$$

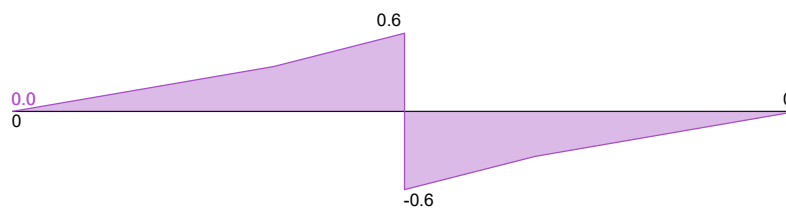
Strip base pressures

$$q_{umax} = \max(q_{u1}, q_{u2}) = 1.092 \text{ ksf}$$

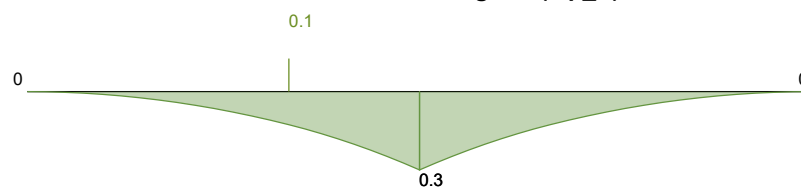
Shear diagram (kips)

Minimum ultimate base pressure

Maximum ultimate base pressure



Moment diagram (kip_ft)



Moment design, y direction, positive moment

Ultimate bending moment

$$M_{u,y,max} = 0.117 \text{ kip_ft}$$

Tension reinforcement provided

No.5 bars at 8.0 in c/c bottom

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Area of tension reinforcement provided

$$A_{sy,bot,prov} = \mathbf{0.465 \text{ in}^2}$$

Minimum area of reinforcement (7.6.1.1)

$$A_{s,min} = 0.0018 \times L_x \times h = \mathbf{0.389 \text{ in}^2}$$

PASS - Area of reinforcement provided exceeds minimum

Maximum spacing of reinforcement (7.7.2.3)

$$s_{max} = \min(3 \times h, 18 \text{ in}) = \mathbf{18 \text{ in}}$$

PASS - Maximum permissible reinforcement spacing exceeds actual spacing

Depth to tension reinforcement

$$d = h - c_{nom,b} - \phi_{y,bot} / 2 = \mathbf{14.688 \text{ in}}$$

Depth of compression block

$$a = A_{sy,bot,prov} \times f_y / (0.85 \times f'_c \times L_x) = \mathbf{0.684 \text{ in}}$$

Neutral axis factor

$$\beta_1 = \mathbf{0.85}$$

Depth to neutral axis

$$c = a / \beta_1 = \mathbf{0.804 \text{ in}}$$

Strain in tensile reinforcement

$$\epsilon_t = 0.003 \times d / c - 0.003 = \mathbf{0.05177}$$

Minimum tensile strain(7.3.3.1)

$$\epsilon_{min} = \epsilon_{ty} + 0.003 = \mathbf{0.00500}$$

PASS - Tensile strain exceeds minimum required

Nominal moment capacity

$$M_n = A_{sy,bot,prov} \times f_y \times (d - a / 2) = \mathbf{33.353 \text{ kip_ft}}$$

Flexural strength reduction factor

$$\phi_f = \min(\max(0.65 + 0.25 \epsilon_t - \epsilon_{ty}) / (0.003), 0.65), 0.9) = \mathbf{0.900}$$

Design moment capacity

$$\phi M_n = \phi_f \times M_n = \mathbf{30.018 \text{ kip_ft}}$$

$$M_{u,y,max} / \phi M_n = \mathbf{0.004}$$

PASS - Design moment capacity exceeds ultimate moment load

One-way shear design, y direction

One-way shear design does not apply. Shear failure plane fall outside extents of foundation.

Shrinkage and temperature requirements, x direction

Reinforcement provided

No.5 bars at 8.0 in c/c bottom

Reinforcement area provided in full width

$$A_{sx,prov} = \mathbf{0.93 \text{ in}^2}$$

Minimum area of reinforcement (24.4.4.3.2)

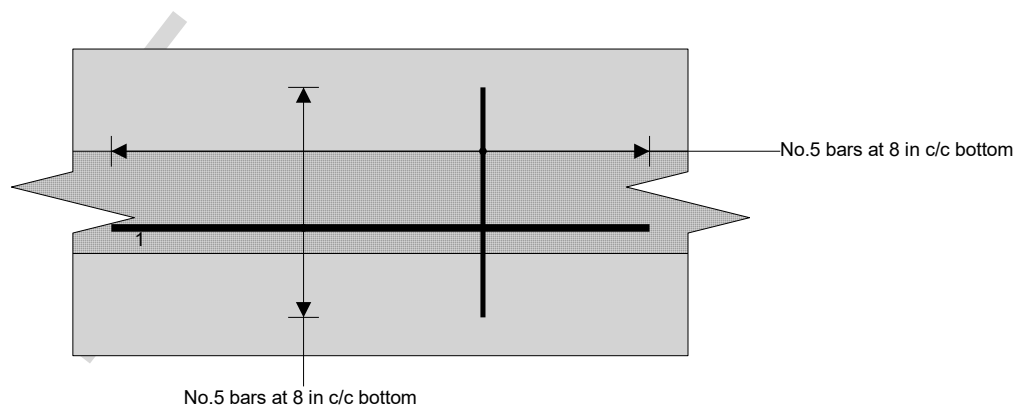
$$A_{s,min,ST} = 0.0018 \times L_y \times h = \mathbf{0.778 \text{ in}^2}$$

PASS - Area of reinforcement provided exceeds minimum

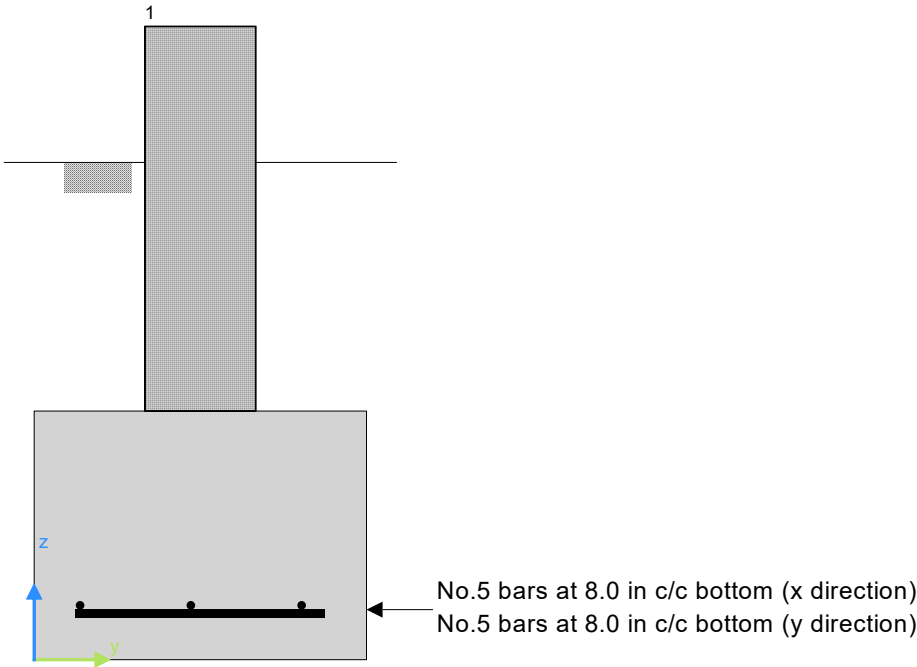
Maximum reinforcement spacing (24.4.4.3.3)

$$s_{max,ST} = \min(5 \times h, 18 \text{ in}) = \mathbf{18 \text{ in}}$$

PASS - Maximum permissible reinforcement spacing exceeds actual spacing



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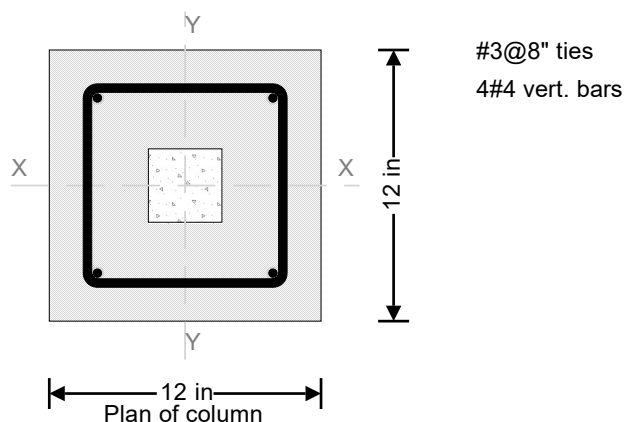
MASONRY COLUMN DESIGN

In accordance with TMS 402/602-16

Tedds calculation version 1.1.04

Summary table

Description	Unit	Provided	Required	Utilization	Result
Under design axial load 3.6 kips:					
Bending	kip_ft	18	1	0.073	PASS
Shear	kips	20	0	0	PASS



Column geometry

Column width	$b = 12 \text{ in}$
Column depth	$d = 12 \text{ in}$
Effective height of column	$h = 16 \text{ ft}$
Gross column area	$A_g = b \times d = 144 \text{ in}^2$
Net column area	$A_n = A_g = 144 \text{ in}^2$
Moment of inertia (x axis)	$I_{gx} = b \times d^3 / 12 = 1728 \text{ in}^4$
Radius of gyration (x axis)	$r_x = \sqrt{I_{gx} / A_g} = 3.46 \text{ in}$
Moment of inertia (y axis)	$I_{gy} = d \times b^3 / 12 = 1728 \text{ in}^4$
Radius of gyration (y axis)	$r_y = \sqrt{I_{gy} / A_g} = 3.46 \text{ in}$
Geometry condition (5.3.1.1)	$h \leq 99 \times \min(r_x, r_y)$

PASS - Column meets dimensional requirements

Masonry unit details

Thickness of masonry unit	$t = 7.625 \text{ in}$
Height of masonry unit	$h_b = 7.625 \text{ in}$
Masonry unit type	Hollow

Masonry details

Pattern bond	Running
Masonry type	Concrete
Density	$\gamma = 115 \text{ lb/ft}^3$
Mortar	PCL
Mortar type	M

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Grout type Coarse
Unit strength $f_{cu} = 1900$ psi
Net area compr. strength (Table 2 TMS 602-16) $f_m = 1900$ psi
Modulus of elasticity (4.2.2) $E_m = 900 \times f_m = 1710000$ psi
Maximum usable strain (9.3.2(c)) $\epsilon_{mu} = 0.0025$

Reinforcement details

Reinforcement yield strength $f_y = 60000$ psi
Modulus of elasticity of reinforcement $E_s = 29000000$ psi
Vertical reinforcement 4 No.4 bars
Minimum clear cover to reinforcement $c_c = 1.5$ in
Area of vertical reinforcement $A_{st} = N_{vert} \times A_{b_vert} = 0.8$ in²
Min. area of vert. reinforcement (5.3.1.3) $A_{st_min} = 0.0025 \times A_n = 0.36$ in²

PASS - Area of vertical reinforcement is greater than minimum required

Max. area of vert. reinforcement (5.3.1.3) $A_{st_max} = 0.04 \times A_n = 5.76$ in²

PASS - Area of vertical reinforcement is less than the maximum allowable

Horizontal tie reinforcement No.3 @ 8 in
Maximum allowable tie spacing (5.3.1.4) $Stie_max = \min(16 \times db_vert, 48 \times db_tie, \min(b,d)) = 8$ in

PASS - Tie spacing is less than maximum allowable spacing

Ultimate loads

Ultimate axial load $P_u = 3.6$ kips
Ultimate shear load $V_u = 0$ kips
Ultimate moment load $M_{ux} = 1.3$ kip_ft

Flexural tensile reinforcement (9.3.3.2)

Ultimate axial load for max. reinf. $P_{u_As} = 3.6$ kips
Tensile strain factor $\alpha = 1.5$
Maximum masonry strain (9.3.2(c)) $\epsilon_{mu} = 0.0025$
Strain in extreme tensile reinforcement $\epsilon_{max} = \alpha \times \epsilon_y = 0.0031$
From strain compatibility analysis
Maximum allowable area of vert. reinf bar $A_{b_vert_max} = 0.63$ in²

PASS - Maximum allowable area of vertical reinforcement exceeds area of vertical reinforcement provided

Check pure axial capacity

Ultimate axial load $P_u = 3.6$ kips
Axial strength reduction factor (9.1.4.4) $\phi = 0.9$
Slenderness ratio $h / r = 55.4$ not greater than 99
Nominal axial design strength (9.3.4.1.1) $P_n = 0.8 \times [0.8 \times f_m \times (A_n - A_{st}) + f_y \times A_{st}] \times [1 - (h / (140 \times r))^2] = 179.2$ kips
Axial design strength $\phi P_n = 161.3$ kips
Axial utilization $P_u / \phi P_n = 0.022$

PASS - Axial design strength exceeds ultimate axial load

Determine moment design strength about x axis under design axial load

Ultimate axial load $P_u = 3.6$ kips
Ultimate moment load $M_{ux} = 1.3$ kip_ft
Depth to extreme tension layer $d_t = 9.87$ in

From strain compatibility analysis:

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Distance to neutral axis, by iteration

Depth of equivalent stress block (9.3.2)

Nominal compression in masonry

Nominal axial force in reinf.

Nominal axial strength

Nominal moment from masonry

Nominal moment in from reinf

Nominal moment strength

Tensile strain in reinforcement

Net tensile strain

Strength reduction factor (9.1.4.4)

Axial design strength, adjusted for slenderness

Bending design strength

Bending utilization

$$c_x = 2.05 \text{ in}$$

$$a = 0.8 \times \min(c_x, d) = 1.64 \text{ in}$$

$$P_{xmas} = 0.8 \times f_m \times (a \times b) = 29.9 \text{ kips}$$

$$P_{xs} = -25.1 \text{ kips}$$

$$P_{nx} = P_{xmas} + P_{xs} = 4.7 \text{ kips}$$

$$M_{xmas} = P_{xmas} \times (d/2 - a/2) = 13 \text{ kip_ft}$$

$$M_{xs} = 7 \text{ kip_ft}$$

$$M_{nx} = M_{xmas} + M_{xs} = 20 \text{ kip_ft}$$

$$\epsilon_t = f_y / E_s = 0.00207$$

$$\epsilon_{tn} = 0.003 \times (d_t - c_x) / c_x = 0.01148$$

$$\phi = 0.9$$

$$\phi P_{nx} = \phi P_{nx} \times [1 - (h / (140 \times r))^2] = 3.6$$

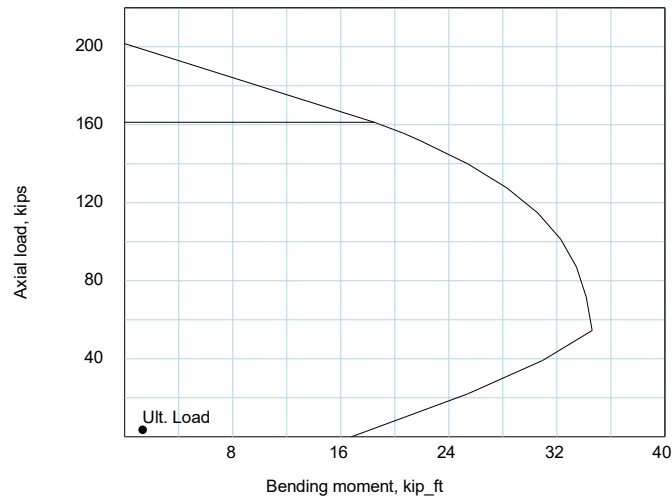
$$\text{kips} \phi M_{nx} = 18 \text{ kip_ft}$$

$$M_{ux} / \phi M_{nx} = 0.073$$

PASS - Design moment strength exceeds ultimate moment load

Check shear capacity

No shear load associated with this load case.



INTERACTION DIAGRAM - X AXIS BENDING ONLY

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FOOTING ANALYSIS

In accordance with ACI318-19 (22)

Tedds calculation version 3.3.14

Summary results

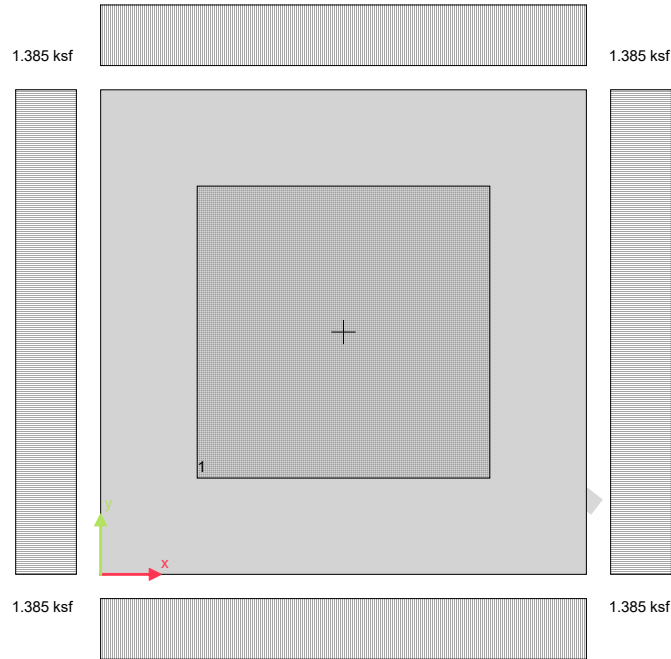
Overall design status **PASS**
Overall design utilization **0.923**

Description	Unit	Applied	Resisting	FoS	Result
Uplift verification	kips	3.8			Pass
Description	Unit	Applied	Resisting	Utilization	Result
Soil bearing	ksf	1.385	1.5	0.923	Pass
Description	Unit	Required	Provided	Utilization	Result
Moment, positive, x-direction	kip_ft	0.1	55.9	0.002	Pass
Moment, positive, y-direction	kip_ft	0.1	52.9	0.002	Pass
Min.area of reinf, bot., x-direction	in ²	0.645	0.880		Pass
Max.reinf.spacing, bot, x-direction	in	18.0	13.1		Pass
Min.area of reinf, bot., y-direction	in ²	0.645	0.880		Pass
Max.reinf.spacing, bot, y-direction	in	18.0	13.1		Pass

Pad footing details

Length of footing $L_x = 1.66$ ft
Width of footing $L_y = 1.66$ ft
Footing area $A = L_x \times L_y = 2.756$ ft²
Depth of footing $h = 18$ in
Depth of soil over footing $h_{soil} = 18$ in
Density of concrete $\gamma_{conc} = 150.0$ lb/ft³

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Column no.1 details

Length of column	$l_{x1} = 12.00 \text{ in}$
Width of column	$l_{y1} = 12.00 \text{ in}$
position in x-axis	$x_1 = 9.96 \text{ in}$
position in y-axis	$y_1 = 9.96 \text{ in}$

Soil properties

Gross allowable bearing pressure	$Q_{allow_Gross} = 1.5 \text{ ksf}$
Density of soil	$\gamma_{soil} = 120.0 \text{ lb/ft}^3$
Angle of internal friction	$\phi_b = 30.0 \text{ deg}$
Design base friction angle	$\delta_{bb} = 30.0 \text{ deg}$
Coefficient of base friction	$\tan(\delta_{bb}) = 0.577$
Self weight	$F_{swt} = h \times \gamma_{conc} = 225 \text{ psf}$
Soil weight	$F_{soil} = h_{soil} \times \gamma_{soil} = 180 \text{ psf}$

Column no.1 loads

Dead load in z	$F_{Dz1} = 2.9 \text{ kips}$
Live roof load in z	$F_{Lrz1} = 1.4 \text{ kips}$

Footing analysis for soil and stability

Load combinations per ASCE 7-16

- 1.0D (0.923)
- 1.0D + 1.0L (0.923)

Combination 1 results: 1.0D

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Forces on footing

Force in z-axis

Moments on footing

Moment in x-axis, about x is 0

$$F_{dz} = \gamma_D \times A \times (F_{swt} + F_{soil}) + \gamma_D \times (F_{Dz1} - l_{x1} \times l_{y1} \times h_{soil} \times \gamma_{soil}) = \mathbf{3.8 \text{ kips}}$$

Moment in y-axis, about y is 0

$$M_{dx} = \gamma_D \times (A \times (F_{swt} + F_{soil}) \times L_x / 2) + \gamma_D \times (((F_{Dz1} - l_{x1} \times l_{y1} \times h_{soil} \times \gamma_{soil})) \times x_1) = \mathbf{3.2 \text{ kip_ft}}$$

Uplift verification

Vertical force

$$M_{dy} = \gamma_D \times (A \times (F_{swt} + F_{soil}) \times L_y / 2) + \gamma_D \times (((F_{Dz1} - l_{x1} \times l_{y1} \times h_{soil} \times \gamma_{soil})) \times y_1) = \mathbf{3.2 \text{ kip_ft}}$$

Bearing resistance

$$F_{dz} = \mathbf{3.816 \text{ kips}}$$

Eccentricity of base reaction

Eccentricity of base reaction in x-axis

Eccentricity of base reaction in y-axis

PASS - Footing is not subject to uplift

Pad base pressures

$$e_{dx} = M_{dx} / F_{dz} - L_x / 2 = \mathbf{0 \text{ in}}$$

$$e_{dy} = M_{dy} / F_{dz} - L_y / 2 = \mathbf{0 \text{ in}}$$

Minimum base pressure

Maximum base pressure

$$q_1 = F_{dz} \times (1 - 6 \times e_{dx} / L_x - 6 \times e_{dy} / L_y) / (L_x \times L_y) = \mathbf{1.385 \text{ ksf}}$$

$$q_2 = F_{dz} \times (1 - 6 \times e_{dx} / L_x + 6 \times e_{dy} / L_y) / (L_x \times L_y) = \mathbf{1.385 \text{ ksf}}$$

$$q_3 = F_{dz} \times (1 + 6 \times e_{dx} / L_x - 6 \times e_{dy} / L_y) / (L_x \times L_y) = \mathbf{1.385 \text{ ksf}}$$

$$q_4 = F_{dz} \times (1 + 6 \times e_{dx} / L_x + 6 \times e_{dy} / L_y) / (L_x \times L_y) = \mathbf{1.385 \text{ ksf}}$$

$$q_{min} = \min(q_1, q_2, q_3, q_4) = \mathbf{1.385 \text{ ksf}}$$

$$q_{max} = \max(q_1, q_2, q_3, q_4) = \mathbf{1.385 \text{ ksf}}$$

Allowable bearing capacity

Allowable bearing capacity

$$q_{allow} = q_{allow_Gross} = \mathbf{1.5 \text{ ksf}}$$

$$q_{max} / q_{allow} = \mathbf{0.923}$$

FOOTING DESIGN

In accordance with ACI318-19 (22)

PASS - Allowable bearing capacity exceeds design base pressure

Material details

Compressive strength of concrete

Yield strength of reinforcement

Tedds calculation version 3.3.14

Compression-controlled strain limit (21.2.2)

Cover to top of footing

Cover to side of footing

Cover to bottom of footing

Concrete type

Concrete modification factor

Column type

$$f'_c = \mathbf{3000 \text{ psi}}$$

$$f_y = \mathbf{60000 \text{ psi}}$$

$$\epsilon_{ty} = \mathbf{0.00200}$$

$$c_{nom_t} = \mathbf{3 \text{ in}}$$

$$c_{nom_s} = \mathbf{3 \text{ in}}$$

$$c_{nom_b} = \mathbf{3 \text{ in}}$$

Normal weight

Analysis and design of concrete footing

Load combinations per ASCE 7-16

1.4D (0.002)

$$\lambda = \mathbf{1.00}$$

Concrete

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1.2D + 1.6L + 0.5Lr (0.002)

Combination 2 results: 1.2D + 1.6L + 0.5Lr

Forces on footing

Ultimate force in z-axis

$$F_{uz} = \gamma_D \times A \times (F_{swt} + F_{soil}) + \gamma_D \times (F_{Dz1} - l_{x1} \times l_{y1} \times h_{soil} \gamma_{soil}) + \gamma_{Lr} \times F_{Lrz1} = 5.3 \text{ kips}$$

Moments on footing

Ultimate moment in x-axis, about x is 0

$$M_{ux} = \gamma_D \times (A \times (F_{swt} + F_{soil}) \times L_x / 2) + \gamma_D \times (((F_{Dz1} - l_{x1} \times l_{y1} \times h_{soil} \gamma_{soil})) \times x_1) + \gamma_{Lr} \times (F_{Lrz1} \times x_1) = 4.4 \text{ kip_ft}$$

$$M_{uy} = \gamma_D \times (A \times (F_{swt} + F_{soil}) \times L_y / 2) + \gamma_D \times (((F_{Dz1} - l_{x1} \times l_{y1} \times h_{soil} \gamma_{soil})) \times y_1) + \gamma_{Lr} \times (F_{Lrz1} \times y_1) = 4.4 \text{ kip_ft}$$

Ultimate moment in y-axis, about y is 0

Eccentricity of base reaction

Eccentricity of base reaction in x-axis

$$e_{ux} = M_{ux} / F_{uz} - L_x / 2 = 0 \text{ in}$$

Eccentricity of base reaction in y-axis

$$e_{uy} = M_{uy} / F_{uz} - L_y / 2 = 0 \text{ in}$$

Pad base pressures

Minimum ultimate base pressure

Maximum ultimate base pressure

$$q_{u1} = F_{uz} \times (1 - 6 \times e_{ux} / L_x - 6 \times e_{uy} / L_y) / (L_x \times L_y) = 1.923 \text{ ksf}$$

$$q_{u2} = F_{uz} \times (1 - 6 \times e_{ux} / L_x + 6 \times e_{uy} / L_y) / (L_x \times L_y) = 1.923 \text{ ksf}$$

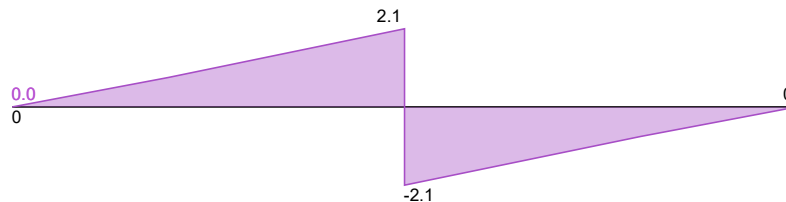
$$q_{u3} = F_{uz} \times (1 + 6 \times e_{ux} / L_x - 6 \times e_{uy} / L_y) / (L_x \times L_y) = 1.923 \text{ ksf}$$

$$q_{u4} = F_{uz} \times (1 + 6 \times e_{ux} / L_x + 6 \times e_{uy} / L_y) / (L_x \times L_y) = 1.923 \text{ ksf}$$

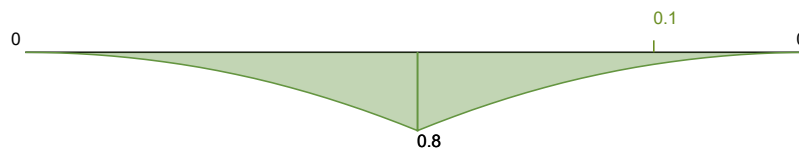
$$q_{umin} = \min(q_{u1}, q_{u2}, q_{u3}, q_{u4}) = 1.923 \text{ ksf}$$

$$q_{umax} = \max(q_{u1}, q_{u2}, q_{u3}, q_{u4}) = 1.923 \text{ ksf}$$

Shear diagram, x axis (kips)



Moment diagram, x axis (kip_ft)



Moment design, x direction, positive moment

Ultimate bending moment

Tension reinforcement provided

Area of tension reinforcement provided

Minimum area of reinforcement (8.6.1.1)

$$M_{u.x.max} = 0.13 \text{ kip_ft}$$

$$2 \text{ No.6 bottom bars (13.1 in c/c)}$$

$$A_{sx.bot.prov} = 0.88 \text{ in}^2$$

$$A_{s.min} = 0.0018 \times L_y \times h = 0.645 \text{ in}^2$$

PASS - Area of reinforcement provided exceeds minimum

Maximum spacing of reinforcement (8.7.2.2)

$$s_{max} = \min(2 \times h, 18 \text{ in}) = 18 \text{ in}$$

PASS - Maximum permissible reinforcement spacing exceeds actual spacing

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Depth to tension reinforcement

$$d = h - c_{nom_b} - \phi_{x.bot} / 2 = 14.625 \text{ in}$$

Depth of compression block

$$a = A_{sx.bot,prov} \times f_y / (0.85 \times f'_c \times L_y) = 1.039 \text{ in}$$

Neutral axis factor

$$\beta_1 = 0.85$$

Depth to neutral axis

$$c = a / \beta_1 = 1.223 \text{ in}$$

Strain in tensile reinforcement

$$\epsilon_t = 0.003 \times d / c - 0.003 = 0.03288$$

Minimum tensile strain(8.3.3.1)

$$\epsilon_{min} = \epsilon_{ty} + 0.003 = 0.00500$$

PASS - Tensile strain exceeds minimum required

Nominal moment capacity

$$M_n = A_{sx.bot,prov} \times f_y \times (d - a / 2) = 62.063 \text{ kip_ft}$$

Flexural strength reduction factor

$$\phi_f = \min(\max(0.65 + 0.25 \epsilon_t - \epsilon_{ty} / (0.003), 0.65), 0.9) = 0.900$$

Design moment capacity

$$\phi M_n = \phi_f \times M_n = 55.857 \text{ kip_ft}$$

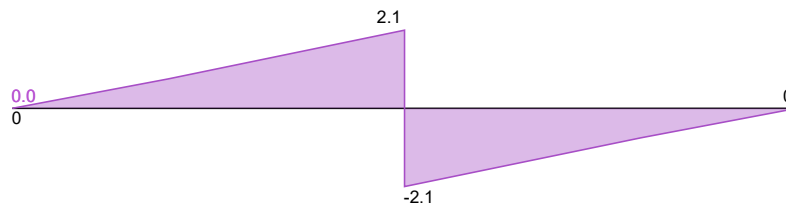
$$M_{u.x,max} / \phi M_n = 0.002$$

PASS - Design moment capacity exceeds ultimate moment load

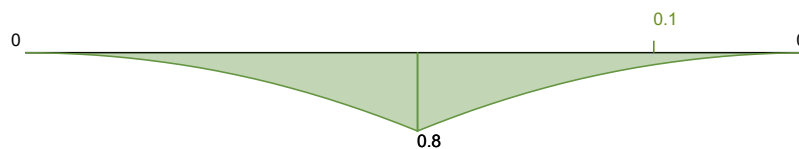
One-way shear design, x direction

One-way shear design does not apply. Shear failure plane fall outside extents of foundation.

Shear diagram, y axis (kips)



Moment diagram, y axis (kip_ft)



Moment design, y direction, positive moment

Ultimate bending moment

$$M_{u.y,max} = 0.13 \text{ kip_ft}$$

Tension reinforcement provided

$$2 \text{ No.6 bottom bars (13.1 in c/c)}$$

Area of tension reinforcement provided

$$A_{sy.bot,prov} = 0.88 \text{ in}^2$$

Minimum area of reinforcement (8.6.1.1)

$$A_{s,min} = 0.0018 \times L_x \times h = 0.645 \text{ in}^2$$

PASS - Area of reinforcement provided exceeds minimum

Maximum spacing of reinforcement (8.7.2.2)

$$s_{max} = \min(2 \times h, 18 \text{ in}) = 18 \text{ in}$$

PASS - Maximum permissible reinforcement spacing exceeds actual spacing

Depth to tension reinforcement

$$d = h - c_{nom_b} - \phi_{x.bot} - \phi_{y.bot} / 2 = 13.875 \text{ in}$$

Depth of compression block

$$a = A_{sy.bot,prov} \times f_y / (0.85 \times f'_c \times L_x) = 1.039 \text{ in}$$

Neutral axis factor

$$\beta_1 = 0.85$$

Depth to neutral axis

$$c = a / \beta_1 = 1.223 \text{ in}$$

Strain in tensile reinforcement

$$\epsilon_t = 0.003 \times d / c - 0.003 = 0.03104$$

Minimum tensile strain(8.3.3.1)

$$\epsilon_{min} = \epsilon_{ty} + 0.003 = 0.00500$$

PASS - Tensile strain exceeds minimum required

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Nominal moment capacity
Flexural strength reduction factor
Design moment capacity

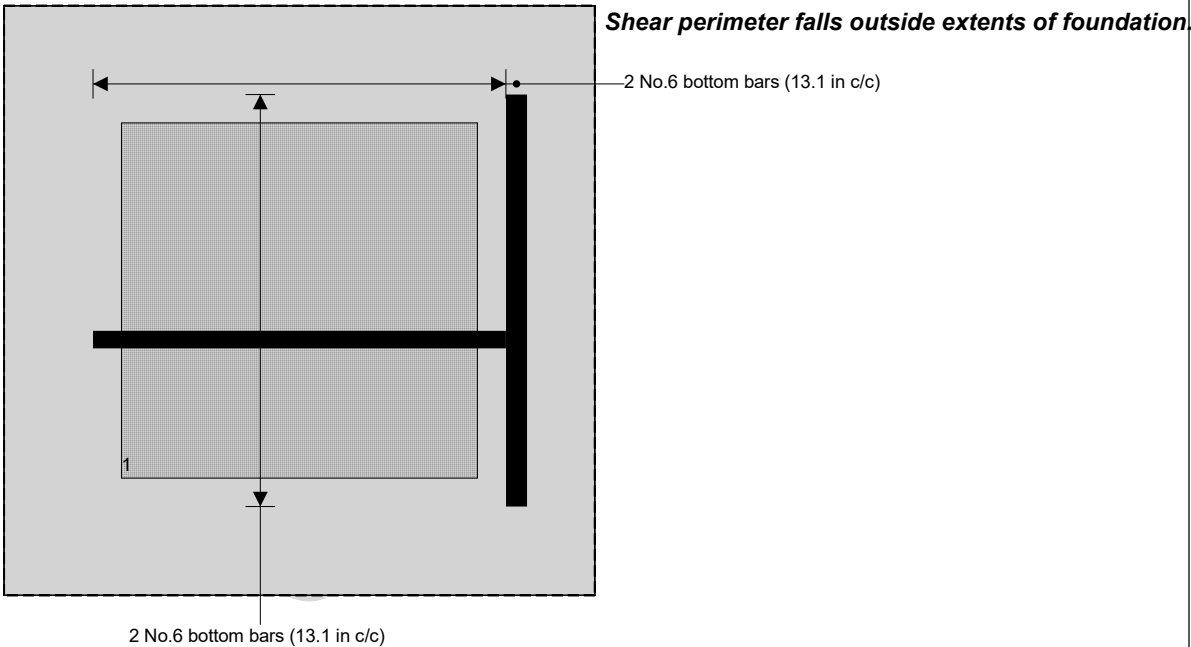
$M_n = A_{sy,bot,prov} \times f_y \times (d - a / 2) = 58.763 \text{ kip_ft}$
 $\phi_f = \min(\max(0.65 + 0.25 \varepsilon_t - \varepsilon_{ty}) / (0.003), 0.65), 0.9) = 0.900$
 $\phi M_n = \phi_f \times M_n = 52.887 \text{ kip_ft}$
 $M_{u,y,max} / \phi M_n = 0.002$

One-way shear design, y direction

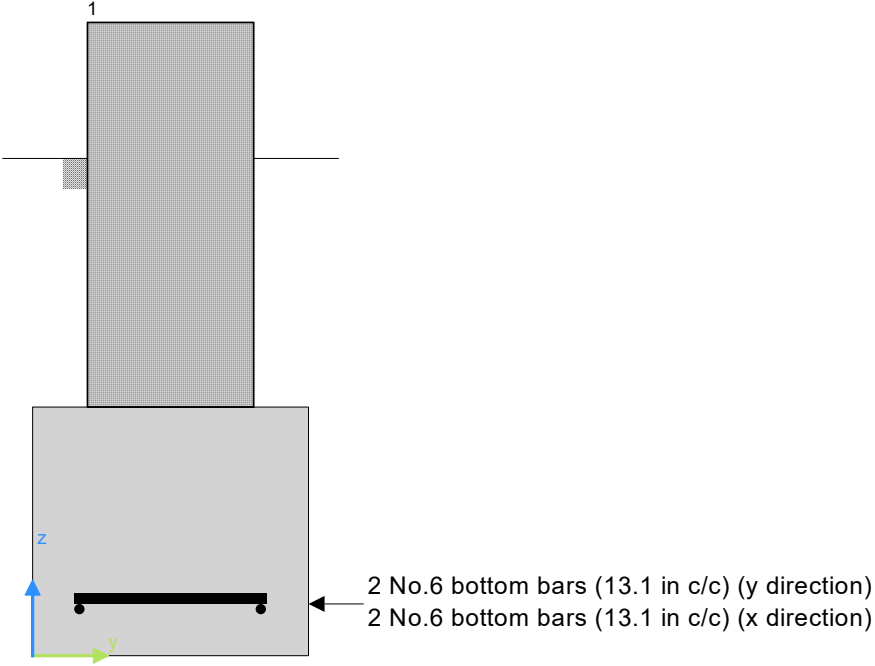
PASS - Design moment capacity exceeds ultimate moment load

One-way shear design does not apply. Shear failure plane fall outside extents of foundation.

Two-way shear design at column 1



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FOOTING ANALYSIS

In accordance with ACI318-19 (22)

Tedds calculation version 3.3.14

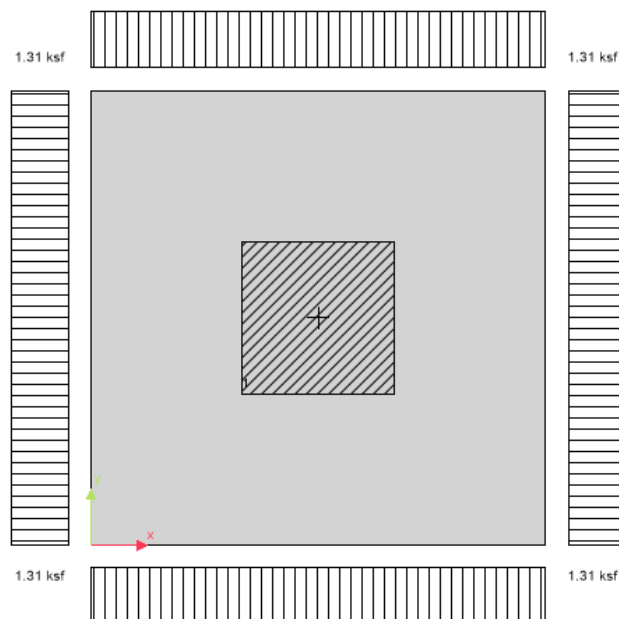
Summary results

Overall design status;

Overall design utilization; 0.873

Pad footing details

Length of footing; $L_x = 1 \text{ ft}$
 Width of footing; $L_y = 1 \text{ ft}$
 Footing area; $A = L_x \times L_y = 1 \text{ ft}^2$
 Depth of footing; $h = 12 \text{ in}$
 Depth of soil over footing; $h_{\text{soil}} = 18 \text{ in}$
 Density of concrete; $\gamma_{\text{conc}} = 150.0 \text{ lb/ft}^3$



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Column no.1 details

Length of column; $l_{x1} = 4.00$ in
Width of column; $l_{y1} = 4.00$ in
position in x-axis; $x_1 = 6.00$ in
position in y-axis; $y_1 = 6.00$ in

Soil properties

Gross allowable bearing pressure; $q_{allow_Gross} = 1.5$ ksf;
Density of soil; $\gamma_{soil} = 120.0$ lb/ft³
Angle of internal friction; $\phi_b = 30.0$ deg
Design base friction angle; $\delta_{bb} = 30.0$ deg
Coefficient of base friction; $\tan(\delta_{bb}) = 0.577$

Footing loads

Self weight; $F_{swt} = h \times \gamma_{conc} = 150$ psf
Soil weight; $F_{soil} = h_{soil} \times \gamma_{soil} = 180$ psf

Column no.1 loads

Dead load in z; $F_{Dz1} = 0.5$ kips
Live load in z; $F_{Lz1} = 0.5$ kips

Footing analysis for soil and stability

Load combinations per ASCE 7-16

1.0D (0.540)
1.0D + 1.0L (0.873)

Combination 2 results: 1.0D + 1.0L

Forces on footing

Force in z-axis; $F_{dz} = \gamma_D \times A \times (F_{swt} + F_{soil}) + \gamma_D \times (F_{Dz1} - l_{x1} \times l_{y1} \times h_{soil} \times \gamma_{soil}) + \gamma_L \times F_{Lz1} = 1.3$ kips

Moments on footing

Moment in x-axis, about x is 0; $M_{dx} = \gamma_D \times (A \times (F_{swt} + F_{soil}) \times L_x / 2) + \gamma_D \times (((F_{Dz1} - l_{x1} \times l_{y1} \times h_{soil} \times \gamma_{soil})) \times x_1) + \gamma_L \times (F_{Lz1} \times x_1) = 0.7$ kip_ft
Moment in y-axis, about y is 0; $M_{dy} = \gamma_D \times (A \times (F_{swt} + F_{soil}) \times L_y / 2) + \gamma_D \times (((F_{Dz1} - l_{x1} \times l_{y1} \times h_{soil} \times \gamma_{soil})) \times y_1) + \gamma_L \times (F_{Lz1} \times y_1) = 0.7$ kip_ft

Uplift verification

Vertical force; $F_{dz} = 1.31$ kips

PASS - Footing is not subject to uplift

Bearing resistance

Eccentricity of base reaction

Eccentricity of base reaction in x-axis; $e_{dx} = M_{dx} / F_{dz} - L_x / 2 = 0$ in
Eccentricity of base reaction in y-axis; $e_{dy} = M_{dy} / F_{dz} - L_y / 2 = 0$ in

Pad base pressures

$q_1 = F_{dz} \times (1 - 6 \times e_{dx} / L_x - 6 \times e_{dy} / L_y) / (L_x \times L_y) = 1.31$ ksf
 $q_2 = F_{dz} \times (1 - 6 \times e_{dx} / L_x + 6 \times e_{dy} / L_y) / (L_x \times L_y) = 1.31$ ksf

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Minimum base pressure;

Maximum base pressure;

Allowable bearing capacity

Allowable bearing capacity;

$$q_3 = F_{dz} \times (1 + 6 \times e_{dx} / L_x - 6 \times e_{dy} / L_y) / (L_x \times L_y) = \mathbf{1.31 \text{ ksf}}$$

$$q_4 = F_{dz} \times (1 + 6 \times e_{dx} / L_x + 6 \times e_{dy} / L_y) / (L_x \times L_y) = \mathbf{1.31 \text{ ksf}}$$

$$q_{\min} = \min(q_1, q_2, q_3, q_4) = \mathbf{1.31 \text{ ksf}}$$

$$q_{\max} = \max(q_1, q_2, q_3, q_4) = \mathbf{1.31 \text{ ksf}}$$

$$q_{\text{allow}} = q_{\text{allow_Gross}} = \mathbf{1.5 \text{ ksf}}$$

$$q_{\max} / q_{\text{allow}} = \mathbf{0.873}$$

PASS - Allowable bearing capacity exceeds design base pressure

FOOTING DESIGN

In accordance with ACI318-19 (22)

Tedds calculation version 3.3.14

Material details

Compressive strength of concrete;

$$f'_c = \mathbf{4000 \text{ psi}}$$

Yield strength of reinforcement;

$$f_y = \mathbf{60000 \text{ psi}}$$

Compression-controlled strain limit (21.2.2);

$$\epsilon_{ty} = \mathbf{0.00200}$$

Cover to top of footing;

$$c_{\text{nom_t}} = \mathbf{3 \text{ in}}$$

Cover to side of footing;

$$c_{\text{nom_s}} = \mathbf{3 \text{ in}}$$

Cover to bottom of footing;

$$c_{\text{nom_b}} = \mathbf{3 \text{ in}}$$

Concrete type;

Normal weight

Concrete modification factor;

$$\lambda = \mathbf{1.00}$$

Column type;

Concrete

Analysis and design of concrete footing

Load combinations per ASCE 7-16

1.4D (0.003)

1.2D + 1.6L + 0.5Lr (0.005)

Combination 2 results: 1.2D + 1.6L + 0.5Lr

Forces on footing

Ultimate force in z-axis;

$$F_{uz} = \gamma_D \times A \times (F_{\text{swt}} + F_{\text{soil}}) + \gamma_D \times (F_{Dz1} - l_{x1} \times l_{y1} \times h_{\text{soil}} \times \gamma_{\text{soil}}) + \gamma_L \times F_{Lz1} = \mathbf{1.8 \text{ kips}}$$

Moments on footing

Ultimate moment in x-axis, about x is 0;

$$M_{ux} = \gamma_D \times (A \times (F_{\text{swt}} + F_{\text{soil}}) \times L_x / 2) + \gamma_D \times (((F_{Dz1} - l_{x1} \times l_{y1} \times h_{\text{soil}} \times \gamma_{\text{soil}})) \times x_1) + \gamma_L \times (F_{Lz1} \times x_1) = \mathbf{0.9 \text{ kip_ft}}$$

Ultimate moment in y-axis, about y is 0;

$$M_{uy} = \gamma_D \times (A \times (F_{\text{swt}} + F_{\text{soil}}) \times L_y / 2) + \gamma_D \times (((F_{Dz1} - l_{x1} \times l_{y1} \times h_{\text{soil}} \times \gamma_{\text{soil}})) \times y_1) + \gamma_L \times (F_{Lz1} \times y_1) = \mathbf{0.9 \text{ kip_ft}}$$

Eccentricity of base reaction

Eccentricity of base reaction in x-axis;

$$e_{ux} = M_{ux} / F_{uz} - L_x / 2 = \mathbf{0 \text{ in}}$$

Eccentricity of base reaction in y-axis;

$$e_{uy} = M_{uy} / F_{uz} - L_y / 2 = \mathbf{0 \text{ in}}$$

Pad base pressures

$$q_{u1} = F_{uz} \times (1 - 6 \times e_{ux} / L_x - 6 \times e_{uy} / L_y) / (L_x \times L_y) = \mathbf{1.772 \text{ ksf}}$$

$$q_{u2} = F_{uz} \times (1 - 6 \times e_{ux} / L_x + 6 \times e_{uy} / L_y) / (L_x \times L_y) = \mathbf{1.772 \text{ ksf}}$$

$$q_{u3} = F_{uz} \times (1 + 6 \times e_{ux} / L_x - 6 \times e_{uy} / L_y) / (L_x \times L_y) = \mathbf{1.772 \text{ ksf}}$$

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Minimum ultimate base pressure;

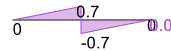
Maximum ultimate base pressure;

$$q_{u4} = F_{uz} \times (1 + 6 \times e_{ux} / L_x + 6 \times e_{uy} / L_y) / (L_x \times L_y) = \mathbf{1.772 \text{ ksf}}$$

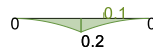
$$q_{umin} = \min(q_{u1}, q_{u2}, q_{u3}, q_{u4}) = \mathbf{1.772 \text{ ksf}}$$

$$q_{umax} = \max(q_{u1}, q_{u2}, q_{u3}, q_{u4}) = \mathbf{1.772 \text{ ksf}}$$

Shear diagram, x axis (kips)



Moment diagram, x axis (kip_ft)



Moment design, x direction, positive moment

Ultimate bending moment;

$$M_{u.x,max} = \mathbf{0.076 \text{ kip_ft}}$$

Tension reinforcement provided;

$$2 \text{ No.4 bottom bars (5.5 in c/c)}$$

Area of tension reinforcement provided;

$$A_{sx,bot,prov} = \mathbf{0.4 \text{ in}^2}$$

Minimum area of reinforcement (8.6.1.1);

$$A_{s,min} = 0.0018 \times L_y \times h = \mathbf{0.259 \text{ in}^2}$$

PASS - Area of reinforcement provided exceeds minimum

Maximum spacing of reinforcement (8.7.2.2);

$$s_{max} = \min(2 \times h, 18 \text{ in}) = \mathbf{18 \text{ in}}$$

PASS - Maximum permissible reinforcement spacing exceeds actual spacing

Depth to tension reinforcement;

$$d = h - C_{nom_b} - \phi_{x.bot} / 2 = \mathbf{8.750 \text{ in}}$$

Depth of compression block;

$$a = A_{sx,bot,prov} \times f_y / (0.85 \times f'_c \times L_y) = \mathbf{0.588 \text{ in}}$$

Neutral axis factor;

$$\beta_1 = \mathbf{0.85}$$

Depth to neutral axis;

$$c = a / \beta_1 = \mathbf{0.692 \text{ in}}$$

Strain in tensile reinforcement;

$$\epsilon_t = 0.003 \times d / c - 0.003 = \mathbf{0.03493}$$

Minimum tensile strain(8.3.3.1);

$$\epsilon_{min} = \epsilon_{ty} + 0.003 = \mathbf{0.00500}$$

PASS - Tensile strain exceeds minimum required

Nominal moment capacity;

$$M_n = A_{sx,bot,prov} \times f_y \times (d - a / 2) = \mathbf{16.912 \text{ kip_ft}}$$

Flexural strength reduction factor;

$$\phi_f = \min(\max(0.65 + 0.25 \times (\epsilon_t - \epsilon_{ty}) / (0.003), 0.65), 0.9) = \mathbf{0.900}$$

Design moment capacity;

$$\phi M_n = \phi_f \times M_n = \mathbf{15.221 \text{ kip_ft}}$$

$$M_{u.x,max} / \phi M_n = \mathbf{0.005}$$

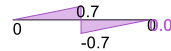
PASS - Design moment capacity exceeds ultimate moment load

One-way shear design, x direction

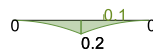
One-way shear design does not apply. Shear failure plane fall outside extents of foundation.

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Shear diagram, y axis (kips)



Moment diagram, y axis (kip_ft)



Moment design, y direction, positive moment

Ultimate bending moment;

$$M_{u,y,max} = 0.076 \text{ kip_ft}$$

Tension reinforcement provided;

2 No.4 bottom bars (5.5 in c/c)

Area of tension reinforcement provided;

$$A_{sy,bot,prov} = 0.4 \text{ in}^2$$

Minimum area of reinforcement (8.6.1.1);

$$A_{s,min} = 0.0018 \times L_x \times h = 0.259 \text{ in}^2$$

PASS - Area of reinforcement provided exceeds minimum

Maximum spacing of reinforcement (8.7.2.2);

$$s_{max} = \min(2 \times h, 18 \text{ in}) = 18 \text{ in}$$

PASS - Maximum permissible reinforcement spacing exceeds actual spacing

Depth to tension reinforcement;

$$d = h - C_{nom_b} - \phi_{x,bot} - \phi_{y,bot} / 2 = 8.250 \text{ in}$$

Depth of compression block;

$$a = A_{sy,bot,prov} \times f_y / (0.85 \times f'_c \times L_x) = 0.588 \text{ in}$$

Neutral axis factor;

$$\beta_1 = 0.85$$

Depth to neutral axis;

$$c = a / \beta_1 = 0.692 \text{ in}$$

Strain in tensile reinforcement;

$$\epsilon_t = 0.003 \times d / c - 0.003 = 0.03276$$

Minimum tensile strain(8.3.3.1);

$$\epsilon_{min} = \epsilon_{ty} + 0.003 = 0.00500$$

PASS - Tensile strain exceeds minimum required

Nominal moment capacity;

$$M_n = A_{sy,bot,prov} \times f_y \times (d - a / 2) = 15.912 \text{ kip_ft}$$

Flexural strength reduction factor;

$$\phi_f = \min(\max(0.65 + 0.25 \times (\epsilon_t - \epsilon_{ty}) / (0.003), 0.65), 0.9) = 0.900$$

Design moment capacity;

$$\phi M_n = \phi_f \times M_n = 14.321 \text{ kip_ft}$$

$$M_{u,y,max} / \phi M_n = 0.005$$

PASS - Design moment capacity exceeds ultimate moment load

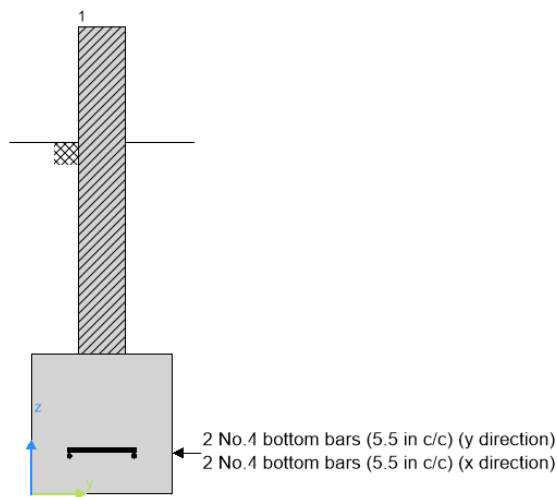
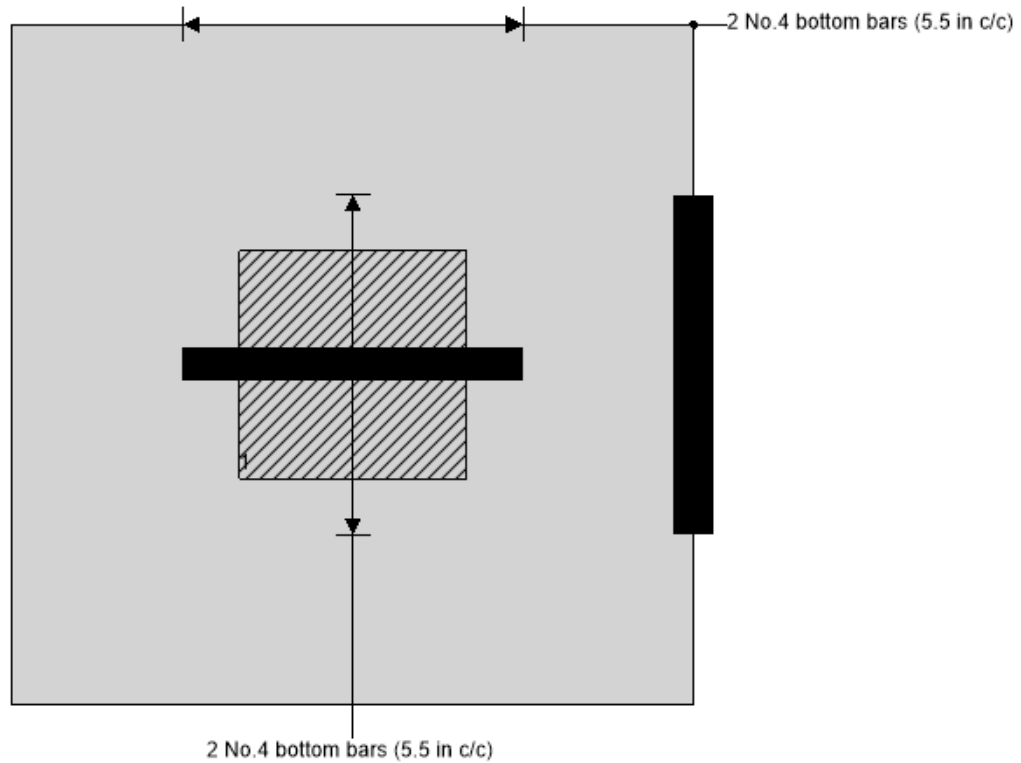
One-way shear design, y direction

One-way shear design does not apply. Shear failure plane fall outside extents of foundation.

Two-way shear design at column 1

Two-way shear design does not apply. Shear perimeter falls outside extents of foundation.

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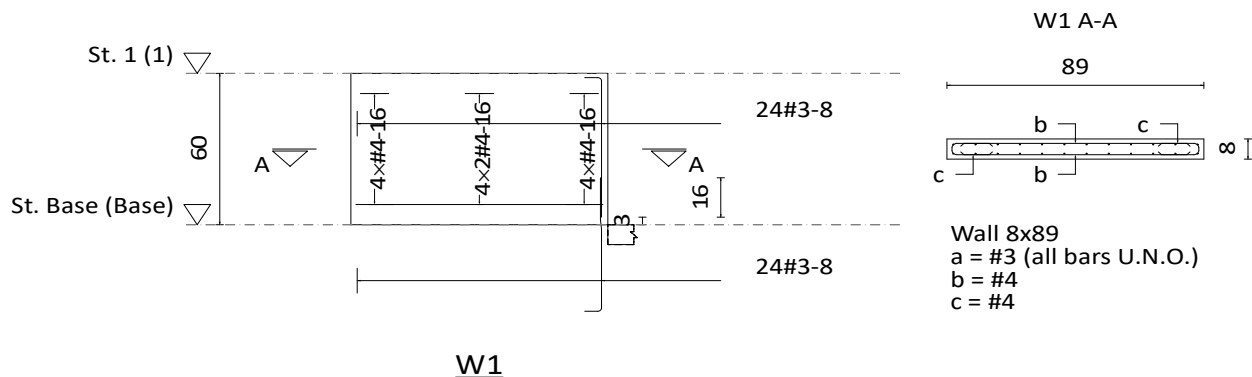


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Concrete**Wall Design**

Walls

W1



Static Design Summary

Vertical Bars Summary

Panel	Length	Thickness	Vertical Bars	Analysis	Combination	Critical position	Ratio	Status
1	7' 5" ft, in	8 in	24#3 - 8	Grillage Chase Down	14	Bottom	0.029	✓ Pass

Shear Summary

Panel	Horizontal Bars	Top support tie legs	Span tie legs	Bottom support tie legs	Analysis	Combination	Ratio	Status
1	#4-16	-	-	-	Grillage Chase Down	16	0.095	✓ Pass

Regional code: United States (ACI/AISC), design code: ACI 318 (2022)

Static Design Calculations

Panel 1 - Critical

Shear - Critical

Grillage Chase Down - Critical

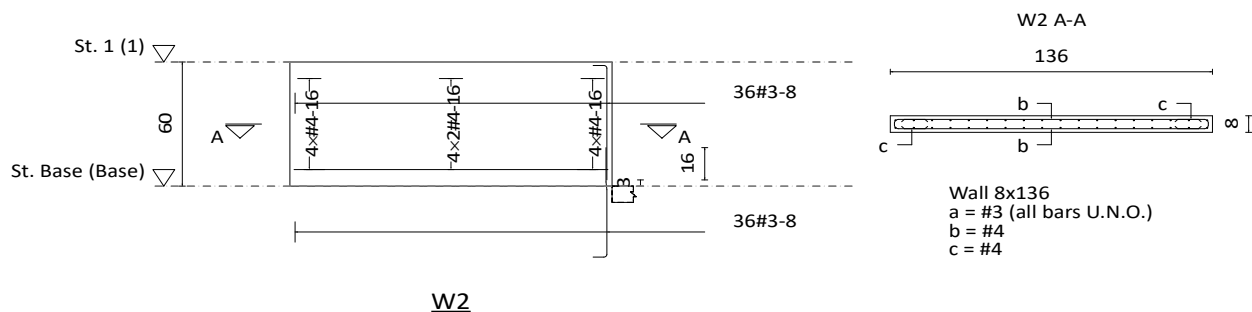
16 LRFD₂-1.2D+1.6L - Critical

Out Of Plane - Critical

 $V_u \leq \phi \times V_c / 2$: unreinforced shear resistance sufficient ACI 318-19(22) Section 10.6.2.1

✓ Pass

W2



Static Design Summary

Vertical Bars Summary

Panel	Length	Thickness	Vertical Bars	Analysis	Combination	Critical position	Ratio	Status
1	11' 4" ft, in	8 in	36#3 - 8	3D Building Analysis	15	Bottom	0.059	✓ Pass

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Shear Summary

Panel	Horizontal Bars	Top support tie legs	Span tie legs	Bottom support tie legs	Analysis	Combination	Ratio	Status
1	#4-16	-	-	-	Grillage Chase Down	15	0.133	✓ Pass

Regional code: United States (ACI/AISC), design code: ACI 318 (2022)

Static Design Calculations

Panel 1 - Critical

Shear - Critical

Grillage Chase Down - Critical

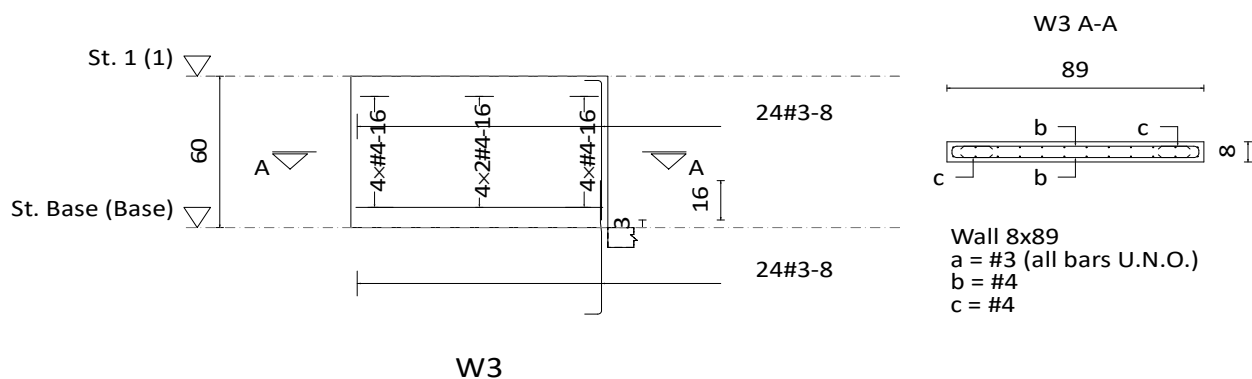
15 LRFD_{1,2}-1.4D+NL_{Dir2+} - Critical

Out Of Plane - Critical

$V_u \leq \phi \times V_c / 2$: unreinforced shear resistance sufficient ACI 318-19(22) Section 10.6.2.1

✓ Pass

W3



Static Design Summary

Vertical Bars Summary

Panel	Length	Thickness	Vertical Bars	Analysis	Combination	Critical position	Ratio	Status
1	7' 5" ft, in	8 in	24#3 - 8	3D Building Analysis	15	Bottom	0.028	✓ Pass

Shear Summary

Panel	Horizontal Bars	Top support tie legs	Span tie legs	Bottom support tie legs	Analysis	Combination	Ratio	Status
1	#4-16	-	-	-	Grillage Chase Down	17	0.088	✓ Pass

Regional code: United States (ACI/AISC), design code: ACI 318 (2022)

Static Design Calculations

Panel 1 - Critical

Shear - Critical

Grillage Chase Down - Critical

17 LRFD_{2,1}-1.2D+1.6L+NL_{Dir1+} - Critical

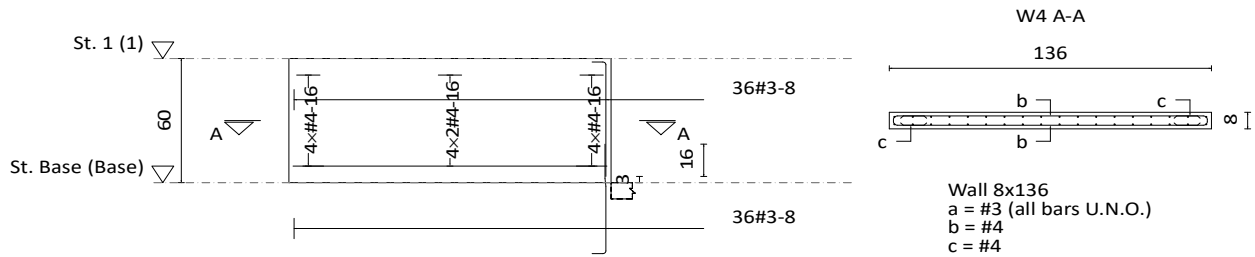
Out Of Plane - Critical

$V_u \leq \phi \times V_c / 2$: unreinforced shear resistance sufficient ACI 318-19(22) Section 10.6.2.1

✓ Pass

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W4



W4

Static Design Summary

Vertical Bars Summary

Panel	Length	Thickness	Vertical Bars	Analysis	Combination	Critical position	Ratio	Status
1	11' 4" ft, in	8 in	36#3 - 8	Grillage Chase Down	13	Bottom	0.060	✓ Pass

Shear Summary

Panel	Horizontal Bars	Top support tie legs	Span tie legs	Bottom support tie legs	Analysis	Combination	Ratio	Status
1	#4-16	-	-	-	Grillage Chase Down	13	0.134	✓ Pass

Regional code: United States (ACI/AISC), design code: ACI 318 (2022)

Static Design Calculations

Panel 1 - Critical

Shear - Critical

Grillage Chase Down - Critical

13 LRFD₁-1.4D - Critical

Out Of Plane - Critical

$V_u \leq \phi \times V_c / 2$: unreinforced shear resistance sufficient ACI 318-19(22) Section 10.6.2.1

✓ Pass

Wall Design Summary

Static

Member Reference	Panel	Grade	Thickness [in]	Utilization	Status
W1	1	3000 psi	8	0.095	✓ Pass
W2	1	3000 psi	8	0.133	✓ Pass
W3	1	3000 psi	8	0.088	✓ Pass
W4	1	3000 psi	8	0.134	✓ Pass

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Concrete

Slab/Mat Design

Slab/Mat Design per Plane

St. Base (Base)

Slab/Mat Panels

Mat: MF 1, Panel: SI 1

Static & RSA Design Summary

Reinforcement Design Summary

Reinforcement Layer	Analysis	Combination	Reinforcement Provided	Area Provided [in ² /ft]	Area Required [in ² /ft]	Utilization
Bottom-X	Grillage Chase Down	16 LRFD ₂ -1.2D+1.6L	#4-13 3/4-B2	0.17	0.01	0.070
Bottom-Y	Grillage Chase Down	16 LRFD ₂ -1.2D+1.6L	#4-13 3/4-B1	0.17	0.01	0.080
Top-X	Grillage Chase Down	16 LRFD ₂ -1.2D+1.6L	#4-13 3/4-T2	0.17	0.01	0.058
Top-Y	3D Building Analysis	18 LRFD _{2,2} -1.2D+1.6L+NL _{Dir2+}	#4-13 3/4-T1	0.17	0.01	0.074

Bearing Pressure Summary

Bearing pressure checks	Max. value [ksf]	Limit [ksf]	Status	Utilization
Pressure	0.57	1.50	✓ Pass	0.378

Slab/Mat Design Summary

Static & RSA

Level	Slab	Reference	Type	Thickness [in]	Grade	Utilization	Status
St. Base (Base) : 0"ft, in	MF 1	SI 1	Foundation mat	7 7/8	3000 psi	0.378	✓ Pass